

【老刘炼芯】十款 Synopsys 芯片设计软件的部署

老刘记事儿 2026-10-02

【摘要】

- 示例了十款 Synopsys 芯片设计软件（VCS、Verdi、Spyglass、DC、Formality、FC、ICC2、PrimeTime、LC、Hspice）在 AlmaLinux 中的详细安装配置过程。
- 展示了 Synopsys License 认证与开机自启动服务在 AlmaLinux 中的配置方法，包含对 SELinux 导致 systemd 服务 203/EXEC 报错的完整排查过程。
- 演示了十款 Synopsys 软件的启动验证方式，可确认软件配置与 License 状态。
- 利用 AI 根据本文说明生成了十款 Synopsys 软件的自动化部署 Bash（Shell）脚本，可快速实现批量软件安装、配置与卸载，支持 AlmaLinux、CentOS 7 和 Ubuntu 系统（已在 CentOS 7.9 中实测确认）。

【引言】

本篇是老刘从零开始自建 ASIC 芯片开发服务器学习全栈芯片开发流程的首篇自学笔记，第一用途是供老刘自己存档经验教训，第二用途是公开出来供有缘人部署参考。因此老刘个人专用的用户名路径等信息文中不作提示请自行甄别替换，背景原因叙述尽量简要直接给结论，读者不明白的可自行搜索确认，也欢迎高手路过批评指教。

老刘已安装配备的是 AlmaLinux 8.10 系统，根据 AI 提示，对于 Synopsys 套件版本，在 AlmaLinux 8 中最推荐 2023 年的版本。这是对 Synopsys 套件版本稳定性、AlmaLinux 系统兼容性、未来 AI ASIC 全自动开发脚本部署的接口成熟性等因素综合考量后的结论。因此老刘决定尽量统一选用 2023.12 作为安装版本。

全栈芯片开发流程先从数字前端设计学起，老刘这里决定先安装下表所列的十款 Synopsys 软件及其安装引导与 License 许可工具，其中除 SCL 选择 2025.03 版本以外，Installer 版本选择为 5.9，其余软件包版本均为 2023.12。

软件简称	安装包	说明
Installer	installer_v5.9	Synopsys 软件安装引导工具
SCL	scl_v2025.03	License 许可工具
VCS	vcs_vV-2023.12-SP1	功能验证仿真工具
Verdi	verdi_vV-2023.12-SP2	调试诊断工具
Spyglass	spyglass_vV-2023.12-SP1	综合前 RTL 静态检查（Lint/CDC/RDC 等）
DC	syn_vV-2023.12-SP3	综合工具（将 RTL 综合成门级网表）
Formality	fm_vV-2023.12-SP3	形式验证工具（综合前后网表一致性）
FC	fusioncompiler_vV-2023.12	从 RTL 到物理版图的全能工具（新型）
ICC2	icc2_vV-2023.12-SP3	由网表生成版图的布局布线工具（传统）
PrimeTime	Prime_vV-2023.12-SP5-1	静态时序最终审查工具
LC	lib_compiler_vV-2023.12-SP3	库编译工具（.lib 编译为 .db）
Hspice	hspice_vV-2023.12-SP2	晶体管级电路仿真工具

对于芯片设计小白，也可阅览上表简单了解各个软件的分工用途。

其中，DC (Design Compiler) 的安装包名称为《syn_vV-2023.12-SP3》。

应优先安装使用 ICC2，如无旧工艺设计需求不必装 ICC；ICC/ICC2 和 DC 的版本年份尽可能对齐以确保数据流兼容。

Fusion Compiler 与 DC 和 ICC2 的功能存在重叠，这是由于 Fusion Compiler 是新一代从 RTL 到物理版图的全能型工具，尤其对 7nm 以下的先进工艺的 PPA 优化具有显著优势，而 DC+ICC2 流程在传统工艺节点上经历过长期验证更为成熟稳定。

如果设计或验证代码中包含 VHDL，就必须得用 VCS-MX。老刘这里暂时先只装 VCS，可支持 Verilog 和 System Verilog。

Synopsys 软件包为商业软件，其最终用户许可协议 (EULA) 明确规定，客户不得复制、转让或分发其许可产品。这也是数字芯片设计入门门槛高的一大因素，如果不依赖单位平台付费，连完整的软件安装包都不易获取，更不要说授权认证了。老刘对 Synopsys 软件的安装使用没有商业目的，纯属学习研究。文中涉及的商业软件请通过官方正规渠道购买或申请正版授权。

本文中**棕色字体**为在 AlmaLinux 系统命令终端中敲入的指令，**蓝色字体**为对文件的编辑内容。

(一) 安装 Synopsys 软件

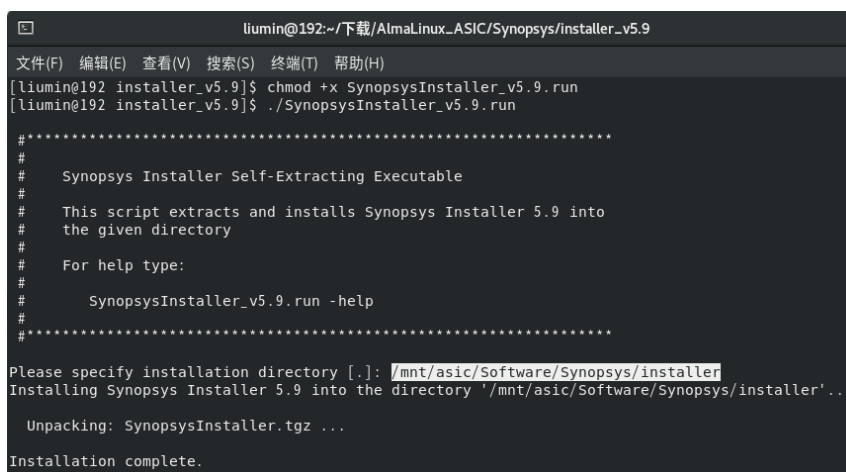
1.1 Synopsys Installer

下载好安装包软件后，现在开始安装，把所有软件安装包拷贝至 AlmaLinux 中的《/home/liumin/下载/AlmaLinux_ASIC/Synopsys/》目录下，而后先装《installer_v5.9》：

```
cd /home/liumin/下载/AlmaLinux_ASIC/Synopsys/installer_v5.9
chmod +x SynopsysInstaller_v5.9.run
./SynopsysInstaller_v5.9.run
```

输入 installer 目标安装位置完成安装：

```
/mnt/asic/Software/Synopsys/installer
```



```
liumin@192:~/下载/AlmaLinux_ASIC/Synopsys/installer_v5.9
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
[liumin@192 installer_v5.9]$ chmod +x SynopsysInstaller_v5.9.run
[liumin@192 installer_v5.9]$ ./SynopsysInstaller_v5.9.run

#*****
#
#   Synopsys Installer Self-Extracting Executable
#
#   This script extracts and installs Synopsys Installer 5.9 into
#   the given directory
#
#   For help type:
#
#       SynopsysInstaller_v5.9.run -help
#*****

Please specify installation directory [.]: /mnt/asic/Software/Synopsys/installer
Installing Synopsys Installer 5.9 into the directory '/mnt/asic/Software/Synopsys/installer'...

Unpacking: SynopsysInstaller.tgz ...
Installation complete.
```

1.2 Synopsys 软件通用安装流程

Synopsys 各软件均可通过 Synopsys Installer 安装，流程高度一致：

(1) 在《~/bashrc》中预先设定环境变量“*****_HOME**”和“**PATH**”，并重加载使之生效：

vim ~/.bashrc

```
export ***_HOME=*****
export PATH=${***_HOME}/bin:$PATH
.....
```

source ~/.bashrc

(2) 切换到 installer 目录下，运行 installer：

cd /mnt/asic/Software/Synopsys/installer
./installer

(3) 输入各软件安装包所在目录地址。

/home/liumin/下载/AlmaLinux_ASIC/Synopsys/****

(4) 输入目标安装路径，首次填入后即可记忆为默认地址，此后安装其他软件时可直接回车确认：**/mnt/asic/Software/Synopsys**

(5) 填写 Site ID number [000]，老刘这里是用默认值，可直接回车确认：**000**

(6) 输入 **yes** 确认，完成该软件安装。

1.3 设定 bashrc

为省去通用安装流程中的第 (1) 步，即每次重复打开《~/bashrc》文件添加设定并重新 source 加载的麻烦，我们可以一次性预填写好该文件中所需的环境变量声明内容。

老刘这里的样例如下：

vim ~/.bashrc

```
# AlmaLinux
export TERM=xterm

# Synopsys License Environment
export Synopsys_DIR=/mnt/asic/Software/Synopsys
export SCL_VERSION=2025.03
export LM_LICENSE_FILE=${Synopsys_DIR}/scl/${SCL_VERSION}/admin/license/Synopsys.dat
export SNPSLMD_LICENSE_FILE=30000@liumin
export SCL_HOME=${Synopsys_DIR}/scl/${SCL_VERSION}/linux64
export PATH=${SCL_HOME}:$PATH
export FLEXLM_DIAGNOSTICS_PATH=${Synopsys_DIR}/scl/${SCL_VERSION}/admin/logs/

# Report Missing Features
export FLEXLM_DIAGNOSTICS=3

# 添加别名方便管理
alias snps-license='${Synopsys_DIR}/scl/${SCL_VERSION}/admin/license/start_license.sh'

# Synopsys VCS
```

```
export VCS_HOME=$Synopsys_DIR/vcs/V-2023.12-SP1
export PATH=$VCS_HOME/bin:$VCS_HOME/amd64/bin:$PATH

# Synopsys Verdi
export VERDI_HOME=$Synopsys_DIR/verdi/V-2023.12-SP2
export PATH=$VERDI_HOME/bin:$PATH

# Synopsys Spyglass
export SPYGLASS_HOME=$Synopsys_DIR/spyglass/V-2023.12-SP1/SPYGLASS_HOME
export PATH=$SPYGLASS_HOME/bin:$PATH

# Synopsys DC
export DC_HOME=$Synopsys_DIR/syn/V-2023.12-SP3
export PATH=$DC_HOME/bin:$PATH

# Synopsys Formality
export FM_HOME=$Synopsys_DIR/fm/V-2023.12-SP3
export PATH=$FM_HOME/bin:$PATH

# Synopsys Fusion Compiler
export FC_HOME=$Synopsys_DIR/fusioncompiler/V-2023.12
export PATH=$FC_HOME/bin:$PATH

# Synopsys ICC2
export ICC2_HOME=$Synopsys_DIR/icc2/V-2023.12-SP3
export PATH=$ICC2_HOME/bin:$PATH
alias icc2='icc2_shell -gui'

# Synopsys PrimeTime
export SYNOPSIS_DGCOM_ROOT=$SCL_HOME
export PT_HOME=$Synopsys_DIR/prime/V-2023.12-SP5-1
export PATH=$PT_HOME/bin:$PATH

# Synopsys Library Compiler
export LC_HOME=$Synopsys_DIR/lc/V-2023.12-SP3
export PATH=$LC_HOME/bin:$PATH

# Synopsys Hspice
export HSPICE_HOME=$Synopsys_DIR/hspice/V-2023.12-SP2/hspice
export PATH=$HSPICE_HOME/bin:$PATH

# Synopsys VIP
export DESIGNWARE_HOME=$Synopsys_DIR/vip/V-2023.12
export PATH=$DESIGNWARE_HOME/bin:$PATH
```

其中：

- (1) “30000@liumin”是 License 服务端口@主机名，主机名须与本机实际 hostname 及 License 文件中登记的取值一致（见 2.1 节）。
- (2) SCL 的 PATH 声明不带“/bin”。
- (3) SPYGLASS_HOME 文件路径与其他软件存在差异，须留意。
- (4) 对于 PrimeTime，须在《~/.bashrc》文件中添加 SYNOPSYS_DGCOM_ROOT、PT_HOME 和 PATH 声明。
- (5) 须留意 HSPICE_HOME 的路径结构与其他软件工具略有不同。
- (6) 关于 Synopsys VIP 的环境变量声明，此处为预填，后续会对 Synopsys VIP 进行专题介绍。

重新加载生效：`source ~/.bashrc`

1.4 SCL 的安装配置

这里首先以 SCL 许可工具的安装作为完整示例，其余软件安装流程与之类似，将只提示安装差异点，不再赘述通用流程部分。

- (1) 因已预设定《~/.bashrc》，此步骤跳过。
- (2) 切换到 installer 目录下，安装 SCL：
`cd /mnt/asic/Software/Synopsys/installer`
`./installer`
- (3) 输入安装包存放地址：`/home/liumin/下载/AlmaLinux_ASIC/Synopsys/scl_v2025.03`
- (4) 输入目标安装路径：`/mnt/asic/Software/Synopsys`
- (5) 填写 Site ID number [000]: `000`
- (6) 输入 `yes` 确认，完成 SCL 安装。

完成 SCL 安装后，建议在安装其他软件之前，先行完成 License 认证（详见 2.1 章节）。以避免在长耗时的软件安装过程完毕后，才发现 License 认证存在异常。

1.5 VCS 的安装配置

只提示差异项：

- (3) 输入安装包存放地址为：
`/home/liumin/下载/AlmaLinux_ASIC/Synopsys/vcs_vV-2023.12-SP1`

输入 `which vcs` 可确认生效：

```
[liumin@192 ~]$ source ~/.bashrc
[liumin@192 ~]$ which vcs
/mnt/asic/Software/Synopsys/vcs/V-2023.12-SP1/bin/vcs
```

1.6 Verdi 的安装配置

只提示差异项：

(3) 输入安装包存放地址：

`/home/liumin/下载/AlmaLinux_ASIC/Synopsys/verdi_vV-2023.12-SP2`

(6) 安装向导会询问是否创建 Verdi 附加组件符号链接，输入 `yes`，回车，`yes` 确认，完成 Verdi 安装。

```
liumin@192:/mnt/asic/Software/Synopsys/installer
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
[liumin@192 installer]$ ./installer

Synopsys (R) Installer
Version 5.9.0

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This software may only be used in accordance with the terms and conditions of a
written license agreement with Synopsys, Inc. All other use, reproduction, or
distribution of this software is strictly prohibited.

Enter the path to the source directory containing the downloaded EFT file(s) [/home/liumin/下载/AlmaLinux_ASIC/
Synopsys/scl_v2021.12]: /home/liumin/下载/AlmaLinux_ASIC/Synopsys/verdi_vV-2023.12-SP2

Enter the full path to the directory where you want to
install Synopsys products. If the directory does not exist,
it will be created. [/mnt/asic/Software/Synopsys]: /mnt/asic/Software/Synopsys
Creating EST file verdi_supp_vV-2023.12-SP2_linux64.spf...
Creating EST file verdi_vV-2023.12-SP2_linux64.spf successfully
Creating EST file verdi_vV-2023.12-SP2_aarch64.spf...
Creating EST file verdi_vV-2023.12-SP2_aarch64.spf successfully
Creating EST file verdi_vV-2023.12-SP2_linux.spf...
Creating EST file verdi_vV-2023.12-SP2_linux.spf successfully
Creating EST file verdi_vV-2023.12-SP2_linux64.spf...
Creating EST file verdi_vV-2023.12-SP2_linux64.spf successfully
Extracting release information from verdi_supp_vV-2023.12-SP2_aarch64.spf
Extracting release information from verdi_supp_vV-2023.12-SP2_aarch64.spf successfully
```

```
liumin@192:/mnt/asic/Software/Synopsys/installer
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
Your site ID number is in the upper-right corner of your Synopsys license
key certificate. If you have trouble locating it, contact your Synopsys
representative.

Site ID number [000]: 000

Post installation setup for VERDI_SUPP:

Create Symbolic Link for Verdi Supplementary Package [Yes]: yes

Please specify VERDI HOME, leave empty for auto detection:

#####
##### Installation Summary #####
Product: VERDI_SUPP (verdi_supp)
Release: V-2023.12-SP2
Platforms: aarch64 common linux linux64
Disk Space: 20128 MB
Target Dir: /mnt/asic/Software/Synopsys/verdi_supp/V-2023.12-SP2

Product: VERDI - The Industry Open Debug Platform (verdi)
Release: V-2023.12-SP2
Platforms: common aarch64 linux linux64
Disk Space: 29862 MB
Target Dir: /mnt/asic/Software/Synopsys/verdi/V-2023.12-SP2

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detecting software piracy and verifying that customers are using Licensed Products in conformity with
the applicable License Key for such Licensed Products. Synopsys will use information gathered in
connection with this process to deliver software updates and pursue software pirates and infringers.
#####
If the information is correct, continue with the installation.
Accept, Install? [yes]: yes
INFO: Prepare for installing V-2023.12-SP2 release of product verdi ...
INFO: Extracting common package from verdi_vV-2023.12-SP2_common.spf ...
INFO: Extracting aarch64 package from verdi_vV-2023.12-SP2_aarch64.spf ...
```

可输入 `which verdi` 确认生效：

```
[liumin@liumin installer]$ which verdi  
/mnt/asic/Software/Synopsys/verdi/V-2023.12-SP2/bin/verdi
```

1.7 Spyglass 的安装配置

只提示差异项：

(3) 输入安装包存放地址：

/home/liumin/下载/AlmaLinux_ASIC/Synopsys/spyglass_vV-2023.12-SP1

需输入 Spyglass Docs 文档包即《spyglass_vV-2023.12-SP1_docs.tar.gz》所在目录位置：

/home/liumin/下载/AlmaLinux_ASIC/Synopsys/spyglass_vV-2023.12-SP1/

可输入 `which spyglass` 确认生效：

```
[liumin@liumin V-2023.12-SP1]$ which spyglass  
/mnt/asic/Software/Synopsys/spyglass/V-2023.12-SP1/SPYGLASS_HOME/bin/spyglass
```

安装完毕后，在《/mnt/asic/Software/Synopsys/》目录下会出现两个安装目录：《spyglass》和《ufe_optional_spyglass-vcs》。

其中 ufe_optional_spyglass-vcs 是 SpyGlass 工具的一个可选功能组件包，它的核心作用是让 SpyGlass 能够调用 VCS 的编译器来处理加密的 RTL 代码。

1.8 DC 的安装配置

只提示差异项：

(3) 输入安装包存放地址：

/home/liumin/下载/AlmaLinux_ASIC/Synopsys/syn_vV-2023.12-SP3

可输入 `which dc_shell` 确认生效：

```
[liumin@liumin ~]$ which dc_shell  
/mnt/asic/Software/Synopsys/syn/V-2023.12-SP3/bin/dc_shell
```

1.9 Formality 的安装配置

只提示差异项：

(3) 输入安装包存放地址：

/home/liumin/下载/AlmaLinux_ASIC/Synopsys/fm_vV-2023.12-SP3

可输入 `which formality` 确认生效：

```
[liumin@liumin ~]$ which formality  
/mnt/asic/Software/Synopsys/fm/V-2023.12-SP3/bin/formality
```


1.10 Fusion Compiler 的安装配置

只提示差异项：

(3) 输入安装包存放地址：

`/home/liumin/下载/AlmaLinux_ASIC/Synopsys/fusioncompiler_vV-2023.12`

可输入 `which fc_shell` 确认生效：

```
[liumin@liumin ~]$ which fc_shell
/mnt/asic/Software/Synopsys/fusioncompiler/V-2023.12/bin/fc_shell
```

1.11 ICC2 的安装配置

只提示差异项：

须先确认 ICC2 的环境依赖库是否齐全，如无则自动补充安装：

`sudo dnf install libnsl libXScrnSaver libXtst libXrender fontconfig expat -y`

(1) 在 1.3 节中已预填的.bashrc 声明中，需要注意如下情况：

对于 AlmaLinux 8.10 系统，由于其内部 xterm-256color terminfo 数据库与 Synopsys 软件的终端初始化库不完全兼容，为避免 ICC2 的 GUI 界面启动时出现“**Information: Term was not able to be set up using xterm-256color . Using "xterm" by default instead. (CLE-10)**”警告信息，需向《~/.bashrc》文件中添加设定：`export TERM=xterm`。

此外，如果不声明 `alias icc2='icc2_shell -gui'`，则启动 ICC2 的图形界面须分为两步，先输入“`icc2_shell`”启动 ICC2 后，再输入“`gui_start`”命令。而声明之后，只需要输入“`icc2`”命令即可启动 ICC2 图形界面。

(3) 输入安装包存放地址：

`/home/liumin/下载/AlmaLinux_ASIC/Synopsys/icc2_vV-2023.12-SP3`

可输入 `which icc2_shell` 确认生效：

```
[liumin@liumin ~]$ which icc2_shell
/mnt/asic/Software/Synopsys/icc2/V-2023.12-SP3/bin/icc2_shell
```

1.12 PrimeTime 的安装配置

只提示差异项：

(3) 输入安装包存放地址：

`/home/liumin/下载/AlmaLinux_ASIC/Synopsys/Prime_vV-2023.12-SP5-1`

(5) 需根据安装提示，提供 SYNOPSYS_DGCOM_ROOT 路径：

`/mnt/asic/Software/Synopsys/scl/2025.03/linux64`

可输入 `which primetime` 或 `which pt_shell` 确认生效：


```
[liumin@liumin ~]$ which primetime
/mnt/asic/Software/Synopsys/prime/V-2023.12-SP5-1/bin/primetime
```

1.13 LC 的安裝配置

只提示差异项：

(3) 输入安装包存放地址：

/home/liumin/下载/AlmaLinux_ASIC/Synopsys/lib_compiler_vV-2023.12-SP3

(6) 安装向导会询问是否工具启动时就检查 License，这里选 **Yes**。

再次输入 **yes** 确认，完成 Library Compiler 安装。

```
Site ID number [000]:

Post installation setup for LC_StandAlone:

  Enable Library Compiler license checkout at tool startup? Yes/No [No]: Yes

#####
##### Installation Summary #####
Product:      LC_StandAlone (lc)
Release:      V-2023.12-SP3
Platforms:    common linux64
Disk Space:   3347 MB
Target Dir:   /mnt/asic/Software/Synopsys/lc/V-2023.12-SP3

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detecting software piracy and verifying that customers are using Licensed Products in conformity with
the applicable License Key for such Licensed Products. Synopsys will use information gathered in
connection with this process to deliver software updates and pursue software pirates and infringers.
#####
If the information is correct, continue with the installation.
Accept, Install? [yes]: yes
INFO: Prepare for installing V-2023.12-SP3 release of product lc ...
INFO: Extracting common package from lc_vV-2023.12-SP3_common.spf ...
INFO: Extracting linux64 package from lc_vV-2023.12-SP3_linux64.spf ...
INFO: Verify SHA256 sum for lc.bz2 ....
INFO: Verify SHA256 sum for linux64/lc.bz2 ....
INFO: Installing V-2023.12-SP3 release of product lc to /mnt/asic/Software/Synopsys/lc/V-2023.12-SP3 ....
INFO: Installing common package for V-2023.12-SP3 release of product lc ...
INFO: Installing linux64 package for V-2023.12-SP3 release of product lc ...
INFO: Running post-installation script: /mnt/asic/Software/Synopsys/lc/V-2023.12-SP3/admin/install/lc/bin/install_
lc_checkout_license -r /mnt/asic/Software/Synopsys/lc/V-2023.12-SP3 -plat "linux64" -checkout "Yes"

Library Compiler Post Installation script running ...

Succeed to generate file /mnt/asic/Software/Synopsys/lc/V-2023.12-SP3/admin/license/.lc_startup_checkout_lic
Library Compiler license checkout is enabled at tool startup.
To change this, set LC_STARTUP_CHECKOUT_LIC to 0 in the user environment before starting a LC session.
```

可输入 **which lc_shell** 确认生效：

```
[liumin@liumin ~]$ which lc_shell
/mnt/asic/Software/Synopsys/lc/V-2023.12-SP3/bin/lc_shell
```

1.14 Hspice 的安裝配置

只提示差异项：

(3) 输入安装包存放地址：

/home/liumin/下载/AlmaLinux_ASIC/Synopsys/hspice_vV-2023.12-SP2

可输入 **which hspice** 确认生效：

```
[liumin@liumin ~]$ which hspice
/mnt/asic/Software/Synopsys/hspice/V-2023.12-SP2/hspice/bin/hspice
```

(二) License 认证与配置

2.1 License 认证

命名 hostname 并确认：

```
sudo hostnamectl set-hostname liumin
hostname
```

```
[liumin@192 installer]$ sudo hostnamectl set-hostname liumin
[sudo] liumin 的密码：
[liumin@192 installer]$ hostname
liumin
```

查看 hostid： `ip addr`

```
[liumin@liumin ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp4s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:14:00:01 d0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.255/24 brd 192.168.1.255 scope global noprefixroute enp4s0
        valid_lft forever preferred_lft forever
    inet6 240e:100:1::1/64 scope global dynamic noprefixroute
        valid_lft 197257sec preferred_lft 110857sec
    inet6 fe80::20c:29ff:fe14:0001 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

如以上两图所示，需将 **hostname (liumin)** 和 **hostid 值 (00*****d0)** 记录下来，用于申请 License 文件授权。

License 文件请通过官方正规渠道或公司授权获得。

老刘的 Synopsys License 文件命名为《Synopsys.dat》。

将《Synopsys.dat》拷贝至 AlmaLinux 环境中的
“/mnt/asic/Software/Synopsys/scl/2025.03/admin/license”目录下。

而后回到 AlmaLinux 环境中，先安装 LSB (Linux Standard Base) 兼容层，以避免运行 Imgrd 时出现“**bash: ./Imgrd: 没有那个文件或目录**”的报错：

```
sudo dnf install redhat-lsb-core -y
```

明确下 License 文件和 Log 文件的目录结构，要有层次规矩，不要乱放：

```

/mnt/asic/Software/Synopsys/scl/2025.03/
├─ admin/
│   └─ license/
│       ├── Synopsys.dat      # License 文件
│       └─ start_license.sh  # 启动脚本
│   └─ logs/
│       └─ snps_license.log  # 日志文件
├─ linux64/
│   └─ bin/
│       ├── lmgrd
│       ├── lmstat
│       └─ lmdown
└─ ...

```

启用 License 服务：

```

cd /mnt/asic/Software/Synopsys/scl/2025.03/linux64/bin
./lmgrd -c /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat -l
/mnt/asic/Software/Synopsys/scl/2025.03/admin/logs/snps_license.log

```

效果如下，SCL License 认证成功。

```

17:26:39 (lmgrd) FlexNet Licensing (v11.19.7.0 build 294770 x64_linux) started on liumin (linux) (8/16/2026)
17:26:39 (lmgrd) Copyright (c) 1988-2024 Flexera. All Rights Reserved.
17:26:39 (lmgrd) World Wide Web: http://www.flexerasoftware.com
17:26:39 (lmgrd) License file(s): /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat
17:26:39 (lmgrd) lmgrd tcp-port 30000
17:26:39 (lmgrd) (@lmgrd-SLOG@) =====
17:26:39 (lmgrd) (@lmgrd-SLOG@) === LMGRD ===
17:26:39 (lmgrd) (@lmgrd-SLOG@) Start-Date: Sun Aug 16 2026 17:26:39 CST
17:26:39 (lmgrd) (@lmgrd-SLOG@) PID: 10762
17:26:39 (lmgrd) (@lmgrd-SLOG@) LMGRD Version: v11.19.7.0 build 294770 x64_linux ( build 294770 (ipv6))
17:26:39 (lmgrd) (@lmgrd-SLOG@)
17:26:39 (lmgrd) (@lmgrd-SLOG@) === Network Info ===
17:26:39 (lmgrd) (@lmgrd-SLOG@) Listening port: 30000
17:26:39 (lmgrd) (@lmgrd-SLOG@)
17:26:39 (lmgrd) (@lmgrd-SLOG@) === Startup Info ===
17:26:39 (lmgrd) (@lmgrd-SLOG@) Server Configuration: Single Server
17:26:39 (lmgrd) (@lmgrd-SLOG@) Command-line options used at LS startup: -c /mnt/asic/Software/Synopsys/scl/2025.03/admin/
license/Synopsys.dat -l /mnt/asic/Software/Synopsys/scl/2025.03/admin/logs/snps_license.log
17:26:39 (lmgrd) (@lmgrd-SLOG@) License file(s) used: /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat
17:26:39 (lmgrd) (@lmgrd-SLOG@) =====
17:26:39 (lmgrd) Starting vendor daemons ...
17:26:39 (lmgrd) Started snpslmd (internet tcp_port 52103 pid 10941)
17:26:39 (snpslmd) FlexNet Licensing version v11.19.7.0 build 294770 x64_linux
08/16/2026 17:26:40 (snpslmd) INFO: Pooling on VENDOR_STRING is Disabled (to enable, set SCL_POOL_VS=1)
08/16/2026 17:26:40 (snpslmd) Synopsys Corporate Licensing (SCL) Release: version SCL_2025.03
08/16/2026 17:26:41 (snpslmd) INFO: License sanity check enabled (360)
08/16/2026 17:26:41 (snpslmd) INFO: File descriptor limit (soft, hard) for snpslmd: 1024, 262144
08/16/2026 17:26:41 (snpslmd) INFO: TCP - tcp_keepalive_time - 7200 | tcp_keepalive_intvl - 75 | tcp_keepalive_probes - 9
| tcp_fin_timeout - 60 | net.ipv4.tcp_max_syn_backlog=256 | net.core.somaxconn=2048
08/16/2026 17:26:41 (snpslmd) INFO: Cloud detection enabled. 1
17:26:41 (snpslmd) SLOG: Summary LOG statistics is enabled.
17:26:41 (snpslmd) SLOG: FNPLS-INTERNAL-CKPT1
17:26:41 (snpslmd) SLOG: VM Status: 0
17:26:41 (snpslmd) SLOG: FNPLS-INTERNAL-CKPT5
17:26:41 (snpslmd) SLOG: TPM Status: 0
17:26:41 (snpslmd) SLOG: FNPLS-INTERNAL-CKPT6

```

中间日志有很多内容……

```
17:26:42 (snpslmd) Server started on liumin for: SSS
17:26:42 (snpslmd) 3D 3P 8alaxy-Zroute
17:26:42 (snpslmd) A3200DX A42MX A500K
17:26:42 (snpslmd) A54SX A54SXA A7
17:26:42 (snpslmd) AARKOS-A ABB_CHCRC_LIBRARY AB_DISABLE_LICENSING
17:26:42 (snpslmd) ACEX1K ACS ACT2-1200XL
17:26:42 (snpslmd) ACT3 AI AIM_DEBUG
17:26:42 (snpslmd) AIM_DEBUG_MEMSSE AIM_DECRYPT AIM_DEVELOPER
17:26:42 (snpslmd) AIM_DEVELOPER_MEMSSE AIM_ENCRYPT ALTGEN1
17:26:42 (snpslmd) ALTGEN2 AN-Impl3D AN-Impl3D_all
17:26:42 (snpslmd) AN-Impl3D_ise AN-Impl3D_ise_all APACM
17:26:42 (snpslmd) APATD APCS APDP
17:26:42 (snpslmd) APEX20K APEX20KC APEX20KE
17:26:42 (snpslmd) APEXII APGA4AP APGACS
17:26:42 (snpslmd) APGADP APGAHP0 APGASolar
17:26:42 (snpslmd) APGASolarII APGATime APGApwr
17:26:42 (snpslmd) APHP0 APPower APRV
17:26:42 (snpslmd) APSolar APSolarII APTIME
17:26:42 (snpslmd) APXtalk APPwr ARKOS-A
17:26:42 (snpslmd) ARKOS-B ARKOS-C ARKOS-D
17:26:42 (snpslmd) ARKOS-E ARKOS-Ice ARKOS-Mcomp
17:26:42 (snpslmd) ARKOS-Rtlcomp ARKOS-Scomp ARKOS-Simul
17:26:42 (snpslmd) ARM7TDMI ATMEL ATMWB_ATMLAYER
17:26:42 (snpslmd) ATMWB_UTOPIA AU AUSEP
17:26:42 (snpslmd) AUT_NUMBER_MAJOR_PACKAGES AdvanceCalibration_all AdvancedCalibration_all
17:26:42 (snpslmd) Apollo ApolloGA Astro
17:26:42 (snpslmd) Astro-Express Astro-MV Astro-Rail
17:26:42 (snpslmd) Astro-Xtalk AstroBeta AstroExp
17:26:42 (snpslmd) AstroGA AstroRail AstroRailSA
17:26:42 (snpslmd) AstroSpecialBeta AstroXtalk BASIC_ANALYSES
17:26:42 (snpslmd) BASIC_ANALYSES_MEMSSE BATCH_MEASURE BATCH_MEASURE_MEMSSE
17:26:42 (snpslmd) BATT_TOOL BATT_TOOL_MEMSSE BC-FPGA
17:26:42 (snpslmd) BC-FPGA-HDL BC-FPGA-VHDL BC-HDL
17:26:42 (snpslmd) BC-Schedule BC-VHDL BETA_LIBRARY
17:26:42 (snpslmd) BOA-BRT BUILD_DATE BUILD_NUMBER
17:26:42 (snpslmd) BUNDLE_MCAD_PROJECTION Behavioral-Analyzer Behavioral-Compiler
```

```
08/16/2026 17:26:42 (snpslmd) Siteid: 26356, Server Hostid: 00-███-███-███ d0, Issued on: 1/8/2018
08/16/2026 17:26:42 (snpslmd) -----
08/16/2026 17:26:42 (snpslmd) INFO: Server WAN NC data enabled
17:26:43 (snpslmd) Changing message security level from -1 to 0
17:26:43 (snpslmd) SLOG: Statistics Log Frequency is 240 minute(s).
17:26:43 (snpslmd) SLOG: TS update poll interval is 0. TS update is detected by midnight reread only.
17:26:43 (snpslmd) SLOG: Activation borrow reclaim percentage is 0.
17:26:43 (snpslmd) (@snpslmd-SLOG@) =====
17:26:43 (snpslmd) (@snpslmd-SLOG@) === Vendor Daemon ===
17:26:43 (snpslmd) (@snpslmd-SLOG@) Vendor daemon: snpslmd
17:26:43 (snpslmd) (@snpslmd-SLOG@) Start-Date: Sun Aug 16 2026 17:26:43 CST
17:26:43 (snpslmd) (@snpslmd-SLOG@) PID: 10941
17:26:43 (snpslmd) (@snpslmd-SLOG@) VD Version: v11.19.7.0 build 294770 x64_linux ( build 294770 (ipv6))
17:26:43 (snpslmd) (@snpslmd-SLOG@)
17:26:43 (snpslmd) (@snpslmd-SLOG@) === Startup/Restart Info ===
17:26:43 (snpslmd) (@snpslmd-SLOG@) Options file used: None
17:26:43 (snpslmd) (@snpslmd-SLOG@) Is vendor daemon a CVD: Yes
17:26:43 (snpslmd) (@snpslmd-SLOG@) Is FlexNet Licensing Service installed and compatible: No
17:26:43 (snpslmd) (@snpslmd-SLOG@) FlexNet Licensing Service Version: -NA-
17:26:43 (snpslmd) (@snpslmd-SLOG@) Is TS accessed: No
17:26:43 (snpslmd) (@snpslmd-SLOG@) TS access time: -NA-
17:26:43 (snpslmd) (@snpslmd-SLOG@) Number of VD restarts since LS startup: 0
17:26:43 (snpslmd) (@snpslmd-SLOG@)
17:26:43 (snpslmd) (@snpslmd-SLOG@) === Network Info ===
17:26:43 (snpslmd) (@snpslmd-SLOG@) Listening port: 52103
17:26:43 (snpslmd) (@snpslmd-SLOG@) Daemon select timeout (in seconds): 1
17:26:43 (snpslmd) (@snpslmd-SLOG@)
17:26:43 (snpslmd) (@snpslmd-SLOG@) === Host Info ===
17:26:43 (snpslmd) (@snpslmd-SLOG@) Host used in license file: liumin
17:26:43 (snpslmd) (@snpslmd-SLOG@) HostID node-locked in license file: 00-███-███-███ d0
17:26:43 (snpslmd) (@snpslmd-SLOG@) HostID of the License Server: 00-███-███-███ d0
17:26:43 (snpslmd) (@snpslmd-SLOG@) Running on Hypervisor: Not determined - treat as Physical
17:26:43 (snpslmd) (@snpslmd-SLOG@) =====
17:26:43 (snpslmd) Ecomms: Encrypted Communication disabled
17:26:43 (snpslmd) TCP_NODELAY NOT enabled
17:26:43 (snpslmd) Listener Thread: running
[liumin@liumin bin]$
```

检查确认 License 状态：

`./lmstat -a -c /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat`

以下响应状态为正常，可打开《snps_license.log》文件进一步检查确认：

```

liumin@liumin:/mnt/asic/Software/Synopsys/scl/2025.03/linux64/bin
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)

[liumin@liumin bin]$ ./lmstat -a -c /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat
lmstat - Copyright (c) 1989-2024 Flexera. All Rights Reserved.
Flexible License Manager status on Fri 10/2/2026 13:18

License server status: 30000@liumin
License file(s) on liumin: /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat:

liumin: license server UP (MASTER) v11.19.7

Vendor daemon status (on liumin):

snpslmd: UP v11.19.7
Feature usage info:

Users of SSS: (Total of 1 license issued; Total of 0 licenses in use)
Users of 3D: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of 3P: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of 8alaxy-Zroute: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of A320DX: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of A42MX: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of A500K: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of A54SX: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of A54SXA: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of A7: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of AARK0S-A: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of ABB_CHCRC_LIBRARY: (Total of 99 licenses issued; Total of 0 licenses in use)
Users of AB_DISABLE_LICENSING: (Total of 99 licenses issued; Total of 0 licenses in use)

```



```
snps_license.log
1000 21:06:35 (snpslmd) gui SYNOPSIS_ANOMALY PrimeTime-New-Technology10
1001 21:06:35 (snpslmd)
1002 21:06:35 (snpslmd) Licenses are case sensitive for TE_CATS
1003 21:06:35 (snpslmd)
1004 21:06:35 (snpslmd) EXTERNAL FILTERS are OFF
1005 21:06:35 (lmgrd) snpslmd using TCP-port 3
1006 21:06:35 (snpslmd) Serving features for the following vendor names:
1007 snpslmd ACAD adalmd anagram arcd atrenta avantd CADABRA chrysalisd coward doradod
EPIC eved extremed fishtail hscd imperasd innologd ISE-TCADD knights magma
metasoftd mwflexd nassd nsysnvs numeritchd pdld perflmd piketec riid saber_dmn
sandwork scilmgrd sftlmd sigmacd SIINT simple snslmgrd ssilmd synforad synopsisd
synplotyd targetld TAVEREN TE_CATS tmalld VALTRIXT vastlmd vcsd wlld
1008 09/26/2026 21:06:35 (snpslmd) -----
1009 09/26/2026 21:06:35 (snpslmd) Checking the integrity of the license file...
1010 09/26/2026 21:06:35 (snpslmd) Valid SSS feature found.
1011 09/26/2026 21:06:35 (snpslmd) The feature is needed to enable the other keys in your license
file.
1012 09/26/2026 21:06:35 (snpslmd) Licensed to liumin
1013 09/26/2026 21:06:35 (snpslmd) Siteid: 26356, Server Hostid: 00000000d0, Issued on: 1/8/2018
1014 09/26/2026 21:06:35 (snpslmd) -----
1015 09/26/2026 21:06:35 (snpslmd) INFO: Server WAN NC data enabled
1016 21:06:36 (snpslmd) Changing message security level from -1 to 0
1017 21:06:36 (snpslmd) SLOG: Statistics Log Frequency is 240 minute(s).
1018 21:06:36 (snpslmd) SLOG: TS update poll interval is 0. TS update is detected by midnight reread
only.
1019 21:06:36 (snpslmd) SLOG: Activation borrow reclaim percentage is 0.
1020 21:06:36 (snpslmd) (@snpslmd-SLOG@) =====
1021 21:06:36 (snpslmd) (@snpslmd-SLOG@) === Vendor Daemon ===
1022 21:06:36 (snpslmd) (@snpslmd-SLOG@) Vendor daemon: snpslmd
1023 21:06:36 (snpslmd) (@snpslmd-SLOG@) Start-Date: Sat Sep 26 2026 21:06:36 CST
1024 21:06:36 (snpslmd) (@snpslmd-SLOG@) PID: 2826
1025 21:06:36 (snpslmd) (@snpslmd-SLOG@) VD Version: v11.19.7.0 build 294770 x64_linux ( build
294770 (ipv6))
1026 21:06:36 (snpslmd) (@snpslmd-SLOG@)
1027 21:06:36 (snpslmd) (@snpslmd-SLOG@) === Startup/Restart Info ===
1028 21:06:36 (snpslmd) (@snpslmd-SLOG@) Options file used: None
1029 21:06:36 (snpslmd) (@snpslmd-SLOG@) Is vendor daemon a CVD: Yes
1030 21:06:36 (snpslmd) (@snpslmd-SLOG@) Is FlexNet Licensing Service installed and compatible: No
1031 21:06:36 (snpslmd) (@snpslmd-SLOG@) FlexNet Licensing Service Version: -NA-
1032 21:06:36 (snpslmd) (@snpslmd-SLOG@) Is TS accessed: No
1033 21:06:36 (snpslmd) (@snpslmd-SLOG@) TS access time: -NA-
1034 21:06:36 (snpslmd) (@snpslmd-SLOG@) Number of VD restarts since LS startup: 0
1035 21:06:36 (snpslmd) (@snpslmd-SLOG@)
1036 21:06:36 (snpslmd) (@snpslmd-SLOG@) === Network Info ===
1037 21:06:36 (snpslmd) (@snpslmd-SLOG@) Listening port: 3
1038 21:06:36 (snpslmd) (@snpslmd-SLOG@) Daemon select timeout (in seconds): 1
1039 21:06:36 (snpslmd) (@snpslmd-SLOG@)
1040 21:06:36 (snpslmd) (@snpslmd-SLOG@) === Host Info ===
1041 21:06:36 (snpslmd) (@snpslmd-SLOG@) Host used in license file: liumin
1042 21:06:36 (snpslmd) (@snpslmd-SLOG@) HostID node-locked in license file: 00000000d0
1043 21:06:36 (snpslmd) (@snpslmd-SLOG@) HostID of the License Server: 00000000d0
1044 21:06:36 (snpslmd) (@snpslmd-SLOG@) Running on Hypervisor: Not determined - treat as Physical
1045 21:06:36 (snpslmd) (@snpslmd-SLOG@) =====
1046 21:06:36 (snpslmd) Ecomms: Encrypted Communication disabled
1047 21:06:36 (snpslmd) TCP_NODELAY NOT enabled
1048 21:06:36 (snpslmd) Listener Thread: running
1049
```

2.2 License 服务开机自启动

2.2.1 编写 SCL License 命令脚本

为了使用方便，得搞个 SCL License 服务的开机自启动脚本。

这分为两步，第一步先编写《start_license.sh》命令脚本：

[vim /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/start_license.sh](#)

粘贴以下内容并保存文件：

```
#!/bin/bash
# Synopsys License Server Startup Script
```

```
SCL_BIN=/mnt/asic/Software/Synopsys/scl/2025.03/linux64/bin
LICENSE_FILE=/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat
LOG_FILE=/mnt/asic/Software/Synopsys/scl/2025.03/admin/logs/snps_license.log

mkdir -p $(dirname ${LOG_FILE})

start() {
    echo "Starting Synopsys License Server..."
    echo "License file: ${LICENSE_FILE}"
    echo "Log file: ${LOG_FILE}"
    ${SCL_BIN}/lmgrd -c ${LICENSE_FILE} -l ${LOG_FILE}
    sleep 2
    ${SCL_BIN}/lmstat -a -c ${LICENSE_FILE}
}

stop() {
    echo "Stopping Synopsys License Server..."
    ${SCL_BIN}/lmdown -c ${LICENSE_FILE} -q
    echo "License server stopped."
}

status() {
    ${SCL_BIN}/lmstat -a -c ${LICENSE_FILE}
}

restart() {
    stop
    sleep 2
    start
}

case "$1" in
    start) start ;;
    stop) stop ;;
    restart) restart ;;
    status) status ;;
    *) echo "Usage: $0 {start|stop|restart|status}"; exit 1 ;;
esac
```

为脚本添加可执行权限：

```
chmod +x /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/start_license.sh
```

测试确认《start_license.sh》脚本运行没问题：


```
./lmdown -c /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat -q  
cd /mnt/asic/Software/Synopsys/scl/2025.03/admin/license  
./start_license.sh start  
./start_license.sh status
```

```
[liumin@liumin license]$ ./start_license.sh status  
lmstat - Copyright (c) 1989-2024 Flexera. All Rights Reserved.  
Flexible License Manager status on Sun 8/16/2026 18:00  
  
License server status: 30000@liumin  
License file(s) on liumin: /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat:  
  
liumin: license server UP (MASTER) v11.19.7  
  
Vendor daemon status (on liumin):  
  
snpslmd: UP v11.19.7  
Feature usage info:  
  
Users of SSS: (Total of 1 license issued; Total of 0 licenses in use)  
Users of 3D: (Total of 99 licenses issued; Total of 0 licenses in use)  
Users of 3P: (Total of 99 licenses issued; Total of 0 licenses in use)  
Users of 8alaxy-Zroute: (Total of 99 licenses issued; Total of 0 licenses in use)  
Users of A3200DX: (Total of 99 licenses issued; Total of 0 licenses in use)
```

2.2.2 创建 systemd 服务实现开机自启动

第二步创建 systemd 服务《snps-license.service》实现开机自启动：

```
sudo vim /etc/systemd/system/snps-license.service
```

粘贴以下内容并保存文件：

```
[Unit]  
Description=Synopsys License Server  
After=network-online.target  
Wants=network-online.target  
  
[Service]  
Type=forking  
User=liumin  
Group=liumin  
  
#增加 10 秒网络依赖启动延时  
ExecStartPre=/bin/sleep 10  
  
ExecStart=/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/start_license.sh start  
ExecStop=/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/start_license.sh stop  
Restart=on-failure  
RestartSec=30  
  
[Install]  
WantedBy=multi-user.target
```

该服务脚本里老刘根据自身应用条件情况，设置了开机 10 秒后的网络依赖启动延时和 30 秒的 systemd 服务重启延时，这是可以根据应用条件变化修订的参数。

而后输入以下指令，分别为：

```
sudo systemctl daemon-reload
sudo systemctl enable snps-license.service
sudo systemctl start snps-license.service
```

执行 snps-license.service 服务状态确认：

```
sudo systemctl status snps-license.service
```

snps-license.service 服务正常运行的状态效果：

```
[liumin@liumin license]$ sudo systemctl status snps-license.service
● snps-license.service - Synopsys License Server
   Loaded: loaded (/etc/systemd/system/snps-license.service; enabled; vendor preset: disabled)
   Active: active (running) since Sun 2026-08-16 18:08:35 CST; 3min 44s ago
     Process: 12843 ExecStart=/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/start_license.sh start (code=exited, status=0/SUCCESS)
     Process: 12841 ExecStartPre=/bin/sleep 10 (code=exited, status=0/SUCCESS)
    Main PID: 12848 (lmgrd)
       Tasks: 2 (limit: 47575)
      Memory: 7.1M
     CGroup: /system.slice/snps-license.service
            └─12848 /mnt/asic/Software/Synopsys/scl/2025.03/linux64/bin/lmgrd -c /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat

8月 16 18:08:35 liumin start_license.sh[12851]: Users of Formality-NewTech1: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin start_license.sh[12851]: Users of Formality-FuSa: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin start_license.sh[12851]: Users of PrimeTime-C-Interface: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin start_license.sh[12851]: Users of checker: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin start_license.sh[12851]: Users of oemunlock: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin start_license.sh[12851]: Users of BasePolicy50: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin start_license.sh[12851]: Users of gui: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin start_license.sh[12851]: Users of SYNOPSYS_ANOMALY: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin start_license.sh[12851]: Users of PrimeTime-New-Technology10: (Total of 99 licenses issued; Total of 0 licenses in use)
8月 16 18:08:35 liumin systemd[1]: Started Synopsys License Server.
```

2.2.3 解除 systemd 服务的 SELinux 运行限制

执行 snps-license.service 服务状态确认时，有可能会发现“status=203/EXEC”报错。这可能是由于 AlmaLinux 默认启用了 SELinux，会严格执行安全策略，使任何未被明确允许的操作都会被阻止，从而导致此类出错。

```
[liumin@liumin bin]$ sudo systemctl status snps-license.service
● snps-license.service - Synopsys License Server
   Loaded: loaded (/etc/systemd/system/snps-license.service; enabled; vendor preset: disabled)
   Active: activating (auto-restart) (Result: exit-code) since Sat 2026-09-26 19:01:02 CST; 16s ago
     Process: 19870 ExecStart=/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/start_license.sh start (code=exited, status=203/EXEC)
     Process: 19844 ExecStartPre=/bin/sleep 10 (code=exited, status=0/SUCCESS)
    Tasks: 0 (limit: 202404)
      Memory: 0B
     CGroup: /system.slice/snps-license.service
```

查看 SELinux 状态：

```
getenforce
```

确认返回值为 Enforcing：

```
[liumin@192 ~]$ getenforce
Enforcing
```

分析审查日志，如果 audit2why 显示 Missing type enforcement (TE) allow rule，则确认是 SELinux 策略问题：

```
sudo ausearch -m avc -ts recent | audit2why
```

如以下指令测试确认能使 snps-license.service 生效，不再出现“status=203/EXEC”报错，也可确定是因 SELinux 因素导致。

```
sudo setenforce 0
sudo systemctl restart snps-license.service
sudo systemctl status snps-license.service
```

确认是 SELinux 因素引起，则可为整个 scl/2025.03 目录树添加规则为“usr_t”，并标记核心二进制文件和启动脚本为“bin_t”，即“可执行文件”类型：

```
sudo semanage fcontext -a -t usr_t "/mnt/asic/Software/Synopsys/scl/2025.03(/.*)?"
2>/dev/null || \
```

```
sudo semanage fcontext -m -t usr_t "/mnt/asic/Software/Synopsys/scl/2025.03(/.*)?"
```

```
sudo semanage fcontext -a -t bin_t
"/mnt/asic/Software/Synopsys/scl/2025.03/linux64/bin/(.*)?" 2>/dev/null || \
```

```
sudo semanage fcontext -m -t bin_t
```

```
"/mnt/asic/Software/Synopsys/scl/2025.03/linux64/bin/(.*)?"
```

```
sudo semanage fcontext -a -t bin_t
"/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/start_license.sh" 2>/dev/null || \
```

```
sudo semanage fcontext -m -t bin_t
```

```
"/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/start_license.sh"
```

重新递归应用新的 SELinux 上下文规则：

```
sudo restorecon -Rv /mnt/asic/Software/Synopsys/scl/2025.03/
```

```
[liumin@192 ~]$ sudo semanage fcontext -a -t usr_t "/mnt/asic/Software/Synopsys/scl/2021.12(/.*)?"
[sudo] liumin 的密码:
[liumin@192 ~]$ sudo restorecon -Rv /mnt/asic/Software/Synopsys/scl/2021.12/
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/install.log from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/admin from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/admin/license from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/admin/license/site_info from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/admin/license/Synopsys_License.dat from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/admin/license/start_license.sh from unconfined_u:object_r:bin_t:s0 to unconfined_u:object_r:usr_t:s0
0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/admin/logs from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/admin/logs/snp_license.log from system_u:object_r:unlabeled_t:s0 to system_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/doc from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/doc/foss from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/doc/foss/SCL_20211108_FOSS.pdf from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
Relabeled /mnt/asic/Software/Synopsys/scl/2021.12/doc/SCL_2021.12_Release_Notes.pdf from unconfined_u:object_r:unlabeled_t:s0 to unconfined_u:object_r:usr_t:s0
```

检查关键文件《Imgrd》的上下文是否正确（应显示 bin_t）：

```
ls -lZ /mnt/asic/Software/Synopsys/scl/2025.03/linux64/bin/Imgrd
```

返回 Enforcing 状态，重新验证确认 snps-license.service 服务使能正常：

```
sudo setenforce 1
sudo systemctl restart snps-license.service
sudo systemctl status snps-license.service
```

这样 Synopsys 软件的 License 认证服务开机自启动设定就完成了，接下来可以逐个测试确认已安装软件的启动运行状态。

（三）Synopsys 软件启动验证

本节任务为测试验证在 AlmaLinux 中各个 Synopsys 软件是否可正常启动，确认有无 License 授权或环境依赖异常。

3.1 VCS 启动验证

验证 VCS 是否安装成功，通常可以采取跑通一个简单仿真示例的方式。VCS 流程输出波形文件还可以传递用于后一道工序 Verdi 的部署验证。

为此，老刘创建了一个《vcs_smoke》测试用例，其结构如下图所示：



《vcs_smoke》测试用例虽小，却也五脏俱全。

《run_smoke.sh》是 VCS 编译运行指令脚本，可以 echo 打印出运行状态信息。

```

run_smoke.sh

1  #!/bin/bash
2  # www.fpga.pw
3  # 2026-10-01
4  # VCS smoke test: verifies VCS license/environment and generates an FSDB waveform.
5  #
6  # 产物目录约定：
7  #   work/   — 编译过程文件（simv_smoke 可执行、simv_smoke.daidir 调试库等）
8  #   result/ — 最终结果文件（nf6_mux2.fsdb 波形、smoke.log 日志）
9  #
10 # Usage:
11 #   cd $PATH/vcs_smoke
12 #   bash run_smoke.sh
13
14 set -e
15
16 # Switch to the directory where this script is located
17 cd "$(dirname "$0")"
18
19 # 每次重跑前清空 work/ 与 result/（二者完全由本脚本生成，保证结果目录干净）
20 rm -rf work result
21 mkdir -p work result
22
23 echo "[SMOKE] Running VCS smoke test in $(pwd)..."
24 echo "[SMOKE] Work dir : ./work    (编译过程文件)"
25 echo "[SMOKE] Result dir: ./result  (最终结果：波形 + 日志)"
26
27 vcs -full64 -sverilog \
28     -kdb -debug_access+all \
29     -f smoke.f \
30     -top tb_nf6_mux2 \
31     -R \
32     -o work/simv_smoke \
33     -l result/smoke.log
34
35 # -R 运行时 FSDB 波形生成在当前目录，移入 result/
36 [ -f nf6_mux2.fsdb ] && mv -f nf6_mux2.fsdb result/
37
38 echo "[SMOKE] VCS compile and simulation completed."
39 echo "[SMOKE] FSDB waveform: result/nf6_mux2.fsdb"
40 echo "[SMOKE] View in Verdi: verdi -ssf ./result/nf6_mux2.fsdb -dbdir ./work/simv_smoke.daidir -top tb_nf6_mux2"
41

```

《smoke.f》是 filelist 文件列表。作为极简测试用例，只包含二选一选择器逻辑《nf6_mux2.v》用于 VCS 启动验证，并为其配套简单的测试激励模块《tb_nf6_mux2.v》。

```
smoke.f

1 // Minimal VCS smoke filelist.
2
3 nf6_mux2.v
4 tb_nf6_mux2.v
5
```

```
nf6_mux2.v
1 //-----
2 // www.fpga.pw
3 //
4 // nf6_mux2.v
5 //
6 // 功能：参数化位宽 2 选 1 多路选择器。
7 // select == 2'b01 时输出 in_0，否则输出 in_1。
8 //
9 //-----
10 `timescale 1ps / 1ps
11
12 module nf6_mux2 #(
13     parameter MUX2_WIDTH = 8
14 ) (
15     input [MUX2_WIDTH-1:0] in_0,
16     input [MUX2_WIDTH-1:0] in_1,
17     output [MUX2_WIDTH-1:0] out,
18     input [1:0] select
19 );
20
21 assign out = (select == 2'b01) ? in_0 : in_1;
22
23 endmodule
24
```

```
tb_nf6_mux2.v
1 //-----
2 // www.fpga.pw
3 // 2026-10-01
4 // Simple testbench for nf6_mux2.
5 // Dumps an FSDb file for waveform viewing in Verdi.
6 //-----
7 `timescale 1ps / 1ps
8
9 module tb_nf6_mux2;
10     parameter MUX2_WIDTH = 8;
11
12     reg [MUX2_WIDTH-1:0] in0;
13     reg [MUX2_WIDTH-1:0] in1;
14     reg [1:0] sel;
15     wire [MUX2_WIDTH-1:0] out;
16
17     nf6_mux2 #(.MUX2_WIDTH(MUX2_WIDTH)) uut (
18         .in_0 (in0),
19         .in_1 (in1),
20         .out (out),
21         .select(sel)
22     );
23
24     initial begin
25         $fsdbDumpfile("nf6_mux2.fsd");
26         $fsdbDumpvars(0, tb_nf6_mux2);
27
28         $display("[TB] Starting nf6_mux2 simple testbench");
29
30         in0 = 8'hAA; in1 = 8'h55; sel = 2'b00; #10;
31         sel = 2'b01; #10;
32         sel = 2'b10; #10;
33         sel = 2'b11; #10;
34         in0 = 8'hFF; in1 = 8'h00; sel = 2'b01; #10;
35         in0 = 8'h0F; in1 = 8'hF0; sel = 2'b10; #10;
36
37         $display("[TB] Simulation finished");
38         $finish;
39     end
40 endmodule
41
```

脚本已设置构建好，只需要输入两条指令即可完成 VCS 编译：

cd /home/liumin/Workspace/LM_Synopsys_Scripts/install/verify/vcs_smoke

bash run_smoke.sh

```
[liumin@liumin ~]$ cd /home/liumin/Workspace/LM_Synopsys_Scripts/install/verify/vcs_smoke
[liumin@liumin vcs_smoke]$ bash run_smoke.sh
[SMOKE] Running VCS smoke test in /home/liumin/Workspace/LM_Synopsys_Scripts/install/verify/vcs_smoke...
[SMOKE] Work dir : ./work (编译过程文件)
[SMOKE] Result dir: ./result (最终结果：波形 + 日志)
Chronologic VCS (TM)
Version V-2023.12-SP1_Full64 -- Fri Oct 2 10:51:27 2026

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[SCL] SCL trace is enabled.
FlexNet Licensing checkout error: No such feature exists.
Feature: VCS-BASE-COMPILE
License path: 30000@liumin:
FlexNet Licensing error:-5,147
license file(s): 30000@liumin
[SCL] 10/02/2026 10:51:27 PID:6081 Client:liumin Checkout denied VCS-BASE-COMPILE 2023.12
[SCL] 10/02/2026 10:51:27 Checking status for feature VCSCompiler_Net
[SCL] 10/02/2026 10:51:27 PID:6081 Client:liumin Server:30000@liumin Checkout succeeded VCSCompiler_Net 2023.12
Parsing design file 'nf6_mux2.v'
Parsing design file 'tb_nf6_mux2.v'
Top Level Modules:
tb_nf6_mux2
TimeScale is 1 ps / 1 ps
Starting vcs inline pass...
```

```
1 module and 0 UDP read.
recompiling module tb_nf6_mux2
[SCL] 10/02/2026 10:51:29 PID:6081 Client:liumin checkin (null)
[SCL] SCL trace is enabled.
FlexNet Licensing checkout error: No such feature exists.
Feature: VCS-BASE-COMPILE
License path: 30000@liumin:
FlexNet Licensing error:-5,147
license file(s): 30000@liumin
[SCL] 10/02/2026 10:51:30 PID:6142 Client:liumin Checkout denied VCS-BASE-COMPILE 2023.12
[SCL] 10/02/2026 10:51:30 Checking status for feature VCSCompiler_Net
[SCL] 10/02/2026 10:51:30 PID:6142 Client:liumin Server:30000@liumin Checkout succeeded VCSCompiler_Net 2023.12
rm -f _cuarc*.so _csrc*.so pre_vcsobj.*.so share_vcsobj.*.so
if [ -x ../work/simv_smoke ]; then chmod a-x ../work/simv_smoke; fi
g++ -o ../work/simv_smoke -rdynamic -Wl,-rpath='$(ORIGIN)/simv_smoke.daidir' -Wl,-rpath=../simv_smoke.daidir -Wl,-rpath=/mnt/asic/Software/Synopsys/vcs/V-2023.12-SP1/linux64/lib -L/mnt/asic/Software/Synopsys/vcs/V-2023.12-SP1/linux64/lib -Wl,-rpath-link=/usr/lib64/libnuma.so.1 -objdump -d _6081_archive.1.so SIM.l.o rmapats_mop.o rmapats.o rmar.nd.o rmar.lvm.o _l.o rmar.lvm.o _l.o -lvirsim -lerrorinf -lsnpsmalloc -lvfs -lvcnew -lsimprofile -luclnactive /mnt/asic/Software/Synopsys/vcs/V-2023.12-SP1/linux64/lib/vcs.tls.o -Wl,-whole-archive -lvcscucli -Wl,-no-whole-archive -vcs_pli_stub.o /mnt/asic/Software/Synopsys/vcs/V-2023.12-SP1/linux64/lib/vcs_save_restore.o /mnt/asic/Software/Synopsys/verdi/V-2023.12-SP2/share/PLI/VCS/LINUX64/pli.a -ldl -lc -lm -lpthread -ldt
../work/simv_smoke up to date
Info: [VCS_SAVE_RESTORE.INFO] ASLR (Address Space Layout Randomization) is detected on the machine. To enable $save functionality, ASLR will be switched off and simv re-executed.
Please use "-no_save" simv switch to avoid re-execution or "-suppress=ASLR_DETECTED.INFO" to suppress this message.
Chronologic VCS simulator copyright 1991-2023
Contains Synopsys proprietary information.
Compiler version V-2023.12-SP1_Full64; Runtime version V-2023.12-SP1_Full64; Oct 2 10:51:2026
[SCL] SCL trace is enabled.
FlexNet Licensing checkout error: No such feature exists.
Feature: VCS-BASE-RUNTIME
License path: 30000@liumin:
FlexNet Licensing error:-5,147
license file(s): 30000@liumin
[SCL] 10/02/2026 10:51:35 PID:6289 Client:liumin Checkout denied VCS-BASE-RUNTIME 2023.12
[SCL] 10/02/2026 10:51:35 Checking status for feature VCSRuntime_Net
[SCL] 10/02/2026 10:51:35 PID:6289 Client:liumin Server:30000@liumin Checkout succeeded VCSRuntime_Net 2023.12
*Verdi* Loading libscore_vcs202312.so
FSDB Dumper for VCS, Release Verdi_V-2023.12-SP2, Linux x86_64/64bit, 05/26/2024
(C) 1996 - 2024 by Synopsys, Inc.
*Verdi* : Create FSDB file 'nf6_mux2.fsd'
*Verdi* : Begin traversing the scope (tb_nf6_mux2), layer (0).
*Verdi* : End of traversing.
[TB] Starting nf6_mux2 simple testbench
[TB] Simulation finished
$finish called from file "tb_nf6_mux2.v", line 38.
$finish at simulation time 60
VCS Simulation Report
Time: 60 ps
CPU Time: 0.160 seconds; Data structure size: 0.0Mb
Fri Oct 2 10:51:35 2026
[SCL] 10/02/2026 10:51:35 PID:6289 Client:liumin checkin (null)
CPU time: 196 seconds to compile + .105 seconds to elab + .166 seconds to link + .178 seconds in simulation
Verdi KDB elaboration done and the database successfully generated: 0 error(s), 0 warning(s)
[SMOKE] VCS compile and simulation completed.
[SMOKE] FSDB waveform: result/nf6_mux2.fsd
```

如以上附图所示，可确认 VCS 正常启动运行，且顺利完成编译生成了《nf6_mux2.fsd》波形文件可用于 Verdi 仿真。

这里的截图报错是老刘特意选取用作说明示例的。从打印 Log 可以看出，截图中 License

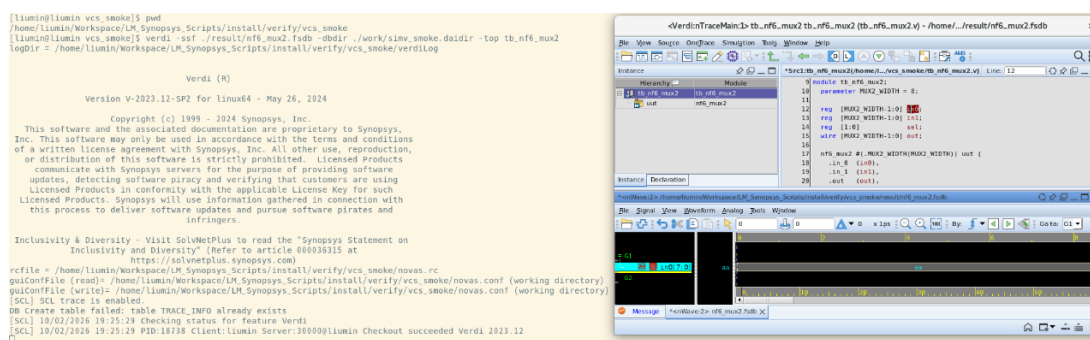
缺乏“VCS-BASE-COMPILE”和“VCS-BASE-RUNTIME”特性，编译运行时会出现 Error 报错，最终是利用“VCSCompiler_Net”和“VCSRuntime_Net”特性完成编译运行流程的。这是由于VCS V-2023.12 支持新旧两种命名体系，会优先尝试签出新命名体系“VCS-BASE-COMPILE”和“VCS-BASE-RUNTIME”，未成功时才会切换并签出旧式 FlexNet 命名体系下的“VCSCompiler_Net”和“VCSRuntime_Net”特性。两套命名体系的编译运行特性本质上并无不同，因此用户对于此类 Error 报错不必担忧。

环节	旧命名 (FlexNet 风格)	新命名 (标准化)
编译	VCSCompiler_Net	VCS-BASE-COMPILE
运行	VCSRuntime_Net	VCS-BASE-RUNTIME

3.2 Verdi 启动验证

有了 VCS 编译运行生成的《nf6_mux2.fsdb》波形文件，可以直接运行以下指令启动 Verdi 仿真界面，确认 Verdi 功能正常可用。

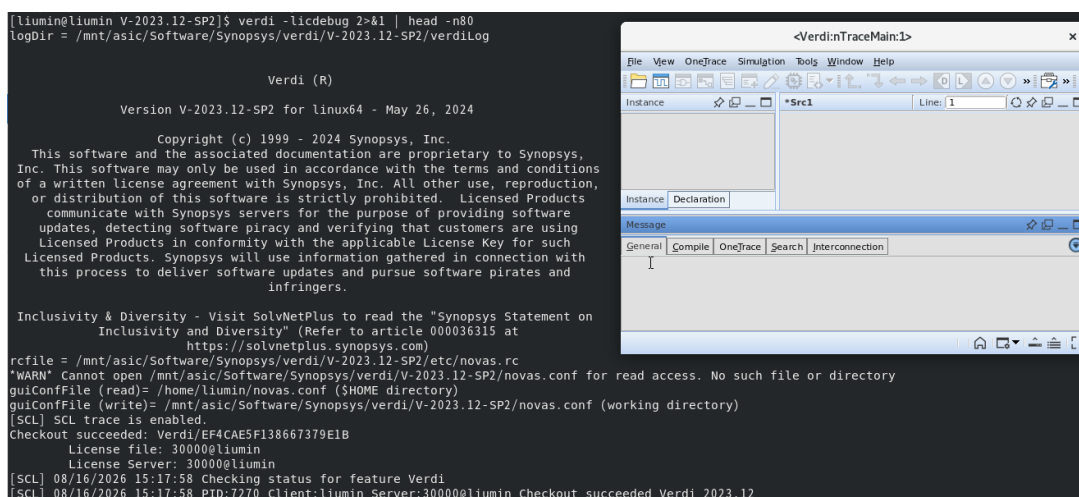
```
verdi -ssf ./result/nf6_mux2.fsdb -dbdir ./work/simv_smoke.daidir -top tb_nf6_mux2
```



在没有前置 VCS 编译运行结果可用的情况下，也可在 License Debug 模式下确认 Verdi 已完成认证可正常启动：

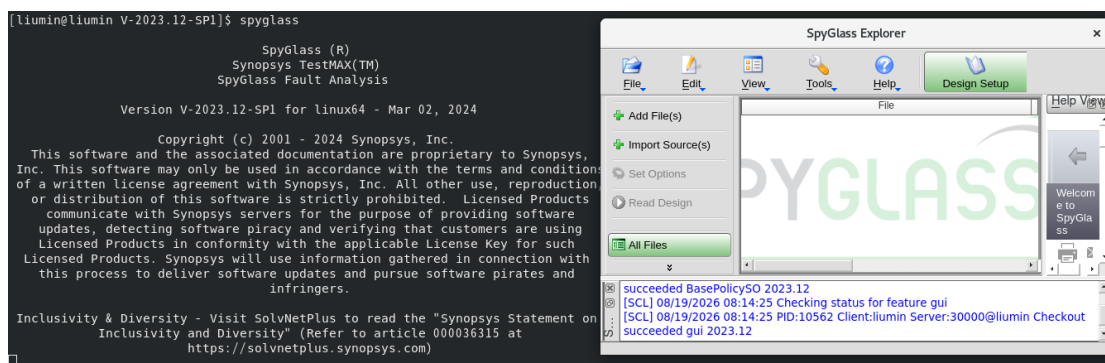
```
verdi -licdebug 2>&1 | head -n80
```

日常使用时也可直接输入 **verdi** 启动软件界面。



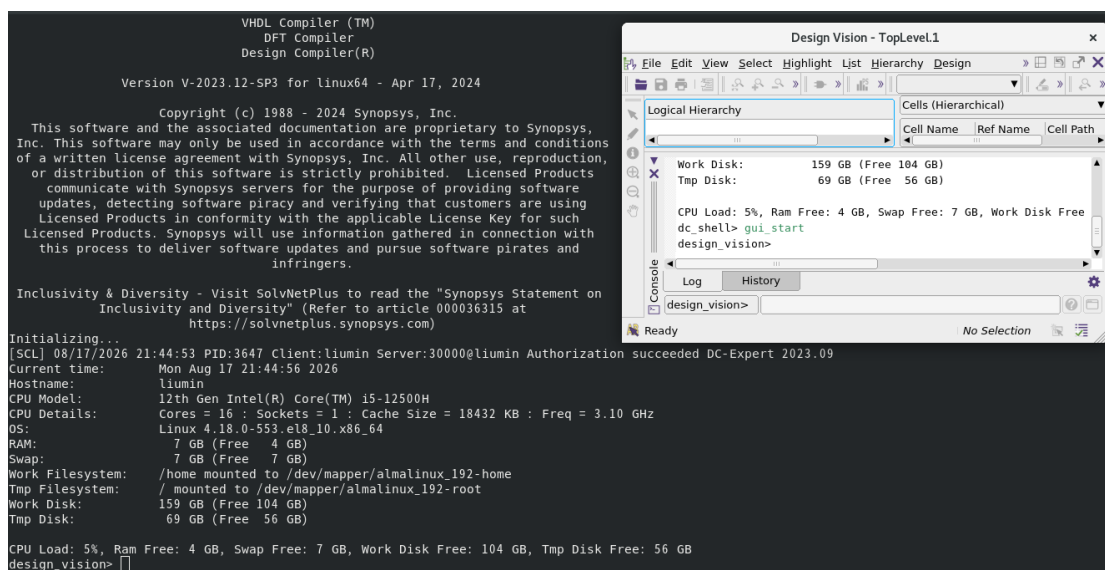
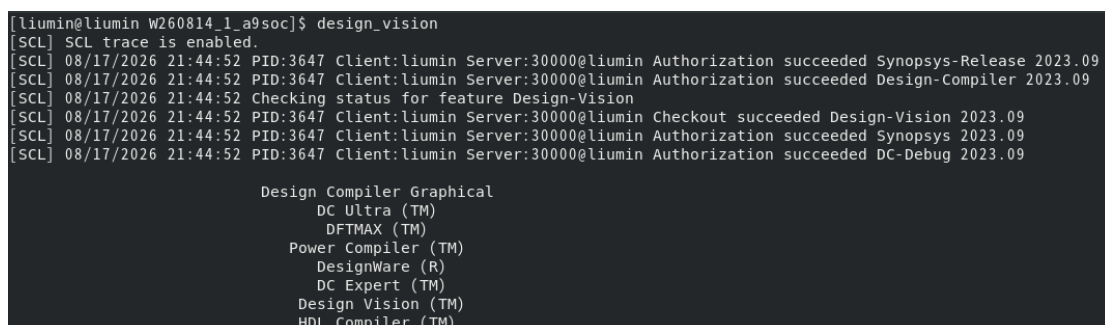
3.3 Spyglass 启动验证

完成认证处理后，输入 **spyglass** 即可启动 Spyglass 软件界面。



3.4 Design Compiler 启动验证

输入 **design_vision** 即可启动 Design Compiler 软件界面。



从 Design Compiler 启动信息来看，老刘的这台 ASIC 开发电脑的 8GB 内存太少了，看起来很容易撑爆。

于是老刘一发狠心从闲鱼花了 730 大洋淘了个长城 32GB DDR4 3200 内存条。

—— 但是后续跑《a9soc》工程的实践证明，32GB 这点内存空间远远不够。

这里可以提前剧透一下，老刘的后续实测表明，在《a9soc》大型 SOC 芯片工程中跑《ddr3_fast_test》仿真，所需占用的内存空间甚至会超出 293GB，这不是普通台式机能承受得了的。

```

Lint-[PCTIO-L] Ports coerced to inout
/home/liumin/Workspace/W260814_1_a9soc/a9soc/design/gpu/src/t622/mali
"cache_ptreqif_req_axiidx"
Port "cache_ptreqif_req_axiidx" declared as input in module
"mali_t622_utlb_ptreqif" may need to be inout. Coercing to inout.

Every 2.0s: free -h                               liumin: Wed Sep  2 21:53:24 2026
Mem:          total      used      free      shared  buff/cache   available
Swap:         263Gi      31Gi      231Gi

```

没办法，在这内存价格上天的时代，想继续玩下去，老刘只能大出血买 ASIC 专用服务器了。具体详情此处不表，看官可有待来篇。

3.5 Formality 启动验证

AlmaLinux 8.10 系统默认缺乏 C Shell, 要使用 Formality 先得安装 csh, 否则启动 Formality 时会出现类似“/bin/csh: 解释器错误：没有那个文件或目录 ERROR: Unknown or unsupported platform”的报错：

```
sudo dnf install tcsh
```

而后输入 **formality** 启动软件界面：

```

[liumin@liumin installer]$ formality
[SC] SCL trace is enabled.
[SC] 08/18/2026 20:32:28 PID:44198 Client:liumin Server:30000@liumin Authorization succeeded Formality 2023.09
[SC] 08/18/2026 20:32:28 Checking status for feature Formality
[SC] 08/18/2026 20:32:28 PID:44198 Client:liumin Server:30000@liumin Checkout succeeded Formality 2023.09

Formality (R)
Version V-2023.12-SP3 for linux64 - Apr 13, 2024

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Build: 8868332
Hostname: liumin
Current time: Tue Aug 18 20:32:28 2026

```

License 中缺乏 Formality-NewTech2 特性出现报错：

```

Loading db file '/mnt/asic/Software/Synopsys/fm/V-2023.12-SP3/libraries/syn/gtech.db'
FlexNet Licensing checkout error: No such feature exists.
Feature: Formality-NewTech2
License path: 30000@liumin:
FlexNet Licensing error:-5,147
license file(s): 30000@liumin
[SC] 08/18/2026 20:32:29 PID:44198 Client:liumin Authorization failed Formality-NewTech2 2023.09
FlexNet Licensing checkout error: No such feature exists.
Feature: Formality-NewTech2
License path: 30000@liumin:
FlexNet Licensing error:-5,147
license file(s): 30000@liumin
[SC] 08/18/2026 20:32:29 PID:44198 Client:liumin Authorization failed Formality-NewTech2 2023.09
Information: Term was not able to be set up using xterm-256color . Using "xterm" by default instead. (CLE-10)

```

检查了下，确实 License 中缺乏这项特性：

```
grep -i "Formality-NewTech2"
/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat
grep -i "Formality"
/mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat
```

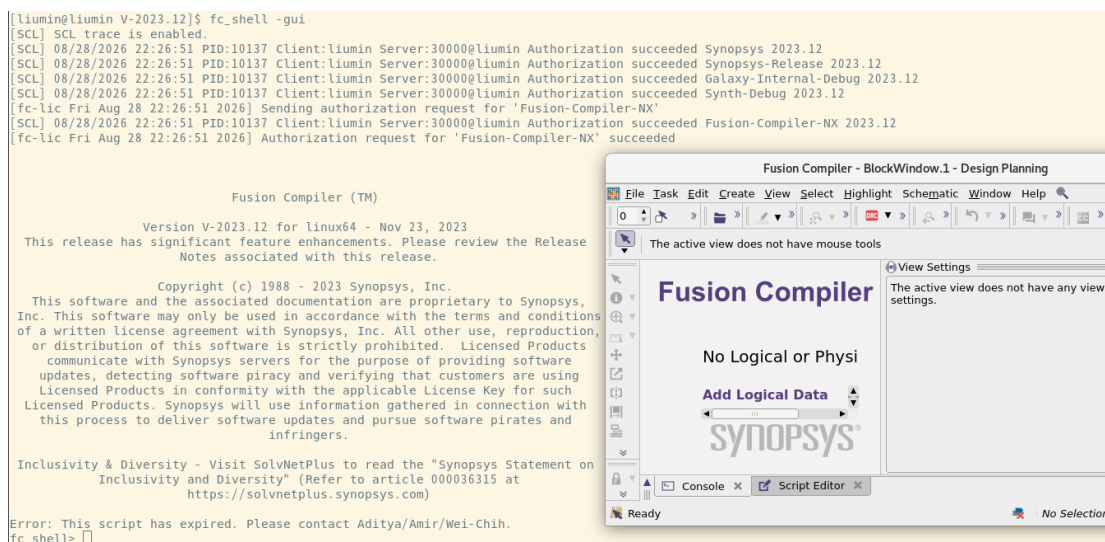
```
[liumin@liumin installer]$ grep -i "Formality-NewTech2" /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat
[liumin@liumin installer]$ grep -i "Formality" /mnt/asic/Software/Synopsys/scl/2025.03/admin/license/Synopsys.dat
INCREMENT Formality snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-Advanced snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-Beta1 snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-Beta2 snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-DV snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-Distributed snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-E1 snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-ECO snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-ESP snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-LogicECO snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-TX snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-TransForm snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-Transit snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-Ultra snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality_advanced_features snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality_basic snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-NewTech1 snpslmd 2030.12 30-dec-2030 99 \
INCREMENT Formality-FuSa snpslmd 2030.12 30-dec-2030 99 \
[liumin@liumin installer]$
```

问了下 AI，原来 Formality-NewTech2 特性是 Formality 工具用来支持较新工艺节点（如 7nm、5nm 及以下）先进优化技术的特定 License Feature。

太高端了，老刘当前阶段顶多玩到 22nm，先不装不要紧。

3.6 Fusion Compiler 启动验证

输入 `fc_shell -gui` 即可启动 Fusion Compiler 软件界面。



```
[liumin@liumin V-2023.12]$ fc_shell -gui
[SCL] SCL trace is enabled.
[SCL] 08/28/2026 22:26:51 PID:10137 Client:liumin Server:30000@liumin Authorization succeeded Synopsys 2023.12
[SCL] 08/28/2026 22:26:51 PID:10137 Client:liumin Server:30000@liumin Authorization succeeded Synopsys-Release 2023.12
[SCL] 08/28/2026 22:26:51 PID:10137 Client:liumin Server:30000@liumin Authorization succeeded Galaxy-Internal-Debug 2023.12
[SCL] 08/28/2026 22:26:51 PID:10137 Client:liumin Server:30000@liumin Authorization succeeded Synth-Debug 2023.12
[fc-lic Fri Aug 28 22:26:51 2026] Sending authorization request for 'Fusion-Compiler-NX'
[SCL] 08/28/2026 22:26:51 PID:10137 Client:liumin Server:30000@liumin Authorization succeeded Fusion-Compiler-NX 2023.12
[fc-lic Fri Aug 28 22:26:51 2026] Authorization request for 'Fusion-Compiler-NX' succeeded

Fusion Compiler (TM)

Version V-2023.12 for linux64 - Nov 23, 2023
This release has significant feature enhancements. Please review the Release
Notes associated with this release.

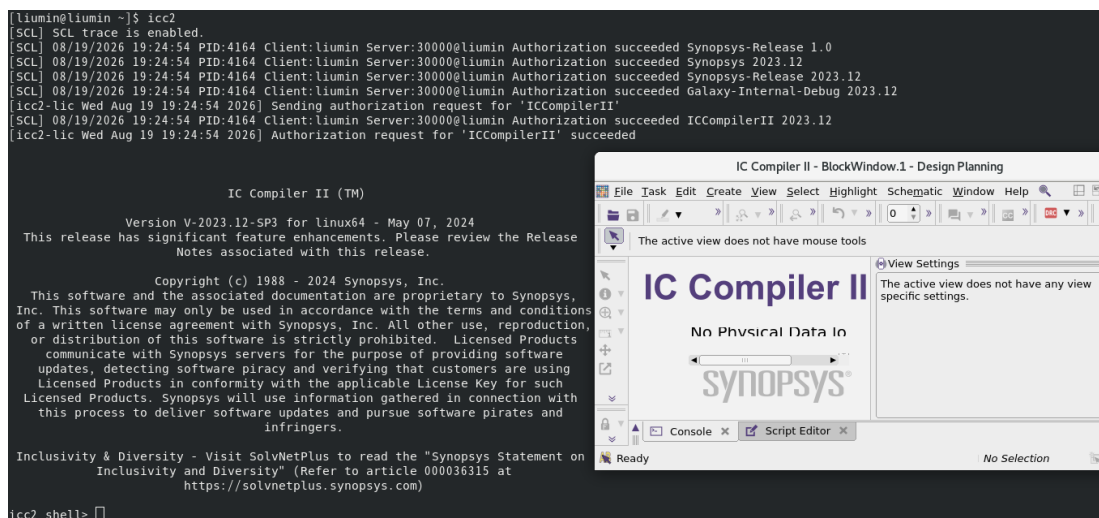
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Error: This script has expired. Please contact Aditya/Amir/Wei-Chih.
fc_shell>
```

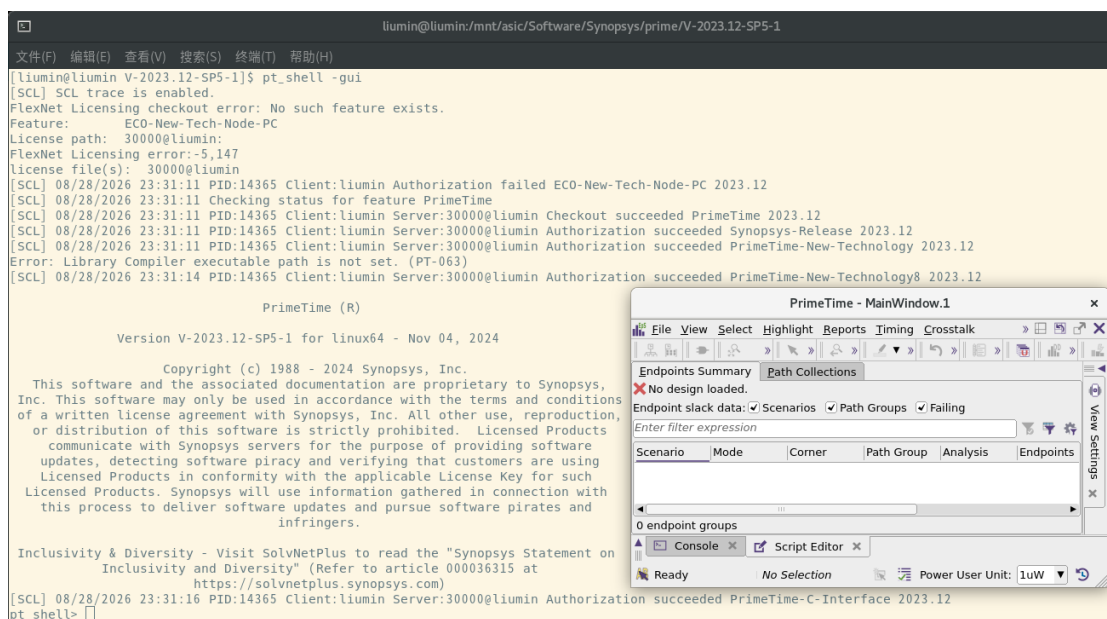
3.7 ICC2 启动验证

由于已在《~/bashrc》中自定义声明，输入 `icc2` 即可启动软件界面：



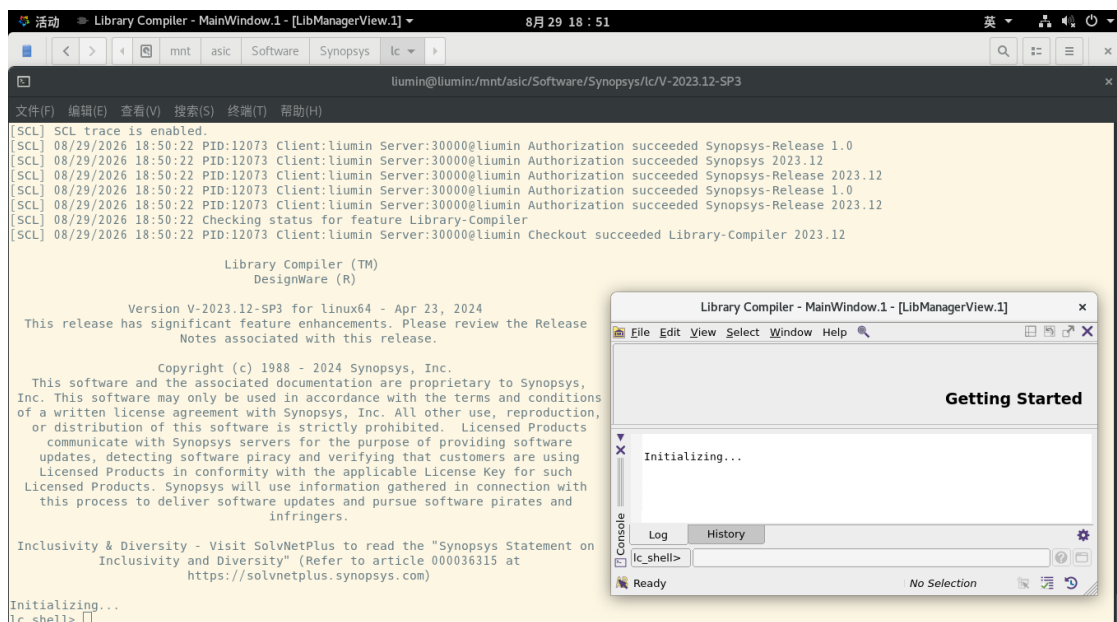
3.8 PrimeTime 启动验证

输入 `pt_shell -gui` 或 `primitime` 即可启动 PrimeTime 软件界面。



3.9 LC 启动验证

输入 `lc_shell -gui` 即可启动 Library Compiler 软件界面。



3.10 Hspice 启动验证

输入 **hspice -l** 即可确认软件安装与 License 配置正常。

```
[liumin@liumin V-2023.12-SP2]$ which hspice
/mnt/asic/Software/Synopsys/hspice/V-2023.12-SP2/hspice/bin/hspice
[liumin@liumin V-2023.12-SP2]$ hspice -I
[SCL] SCL trace is enabled.
lic:
lic: FLEXlm: SDK_12.11.9.5
lic: USER: liumin          HOSTNAME: liumin
lic: HOSTID: ████████-██████-██01   PID: 14622
[SCL] 08/29/2026 19:23:27 Checking status for feature hspice
[SCL] 08/29/2026 19:23:27 PID:14622 Client:liumin Server:30000@liumin Checkout succeeded hspice 2023.12
[SCL] 08/29/2026 19:23:27 Checking status for feature hspice
lic: Using FLEXlm license file:
lic: 30000@liumin
lic: Checkout 1 hspice
lic: License/Maintenance for hspice will expire on 30-dec-2030/2030.12
lic: 1(in_use)/99(total) FLOATING license(s) on SERVER 30000@liumin
lic:
Hspice license have checked out
HSPICE >
```

（四）AI 生成自动化部署脚本

其实这类软件安装部署的活儿非常适合 AI 同学来干，而非人工处理。老刘坚持软件的首次安装部署由自己完成并保留详尽记录，只是因为在这一过程中可以更熟悉并自定义软件部署的层次结构与风格，了解 AlmaLinux 的环境特性。

在将带有个人风格喜好的经验教训说明（本文）作为自定义安装规范喂给 AI 后，再次部署就可以完全由 AI 所生成的脚本全自主完成本文内容，并且是能合乎个人心意、能高效达成目标、能节省 Token 避免走弯路的。

在豆包 AI 的帮助下（你没看错，就是那个爱戴红围脖平时嬉皮笑脸的豆包小妹），老刘生成了本文内容所对应的可适用于 AlmaLinux、CentOS 7 和 Ubuntu 三类 Linux 系统的 Synopsys 软件自动化安装 Bash（Shell）脚本。


```

1  # Synopsys 芯片设计软件自动化安装 / 启动 / License 校验脚本
2
3  依据《【老刘炼芯】十款Synopsys芯片设计软件的部署》一文编写。
4  环境基线：AlmaLinux 8.10：Installer 5.9, SCL 2025.03, 其余软件包 2023.12。
5  **跨发行版支持**：同一套脚本可在 AlmaLinux 8 / CentOS 7 / Ubuntu 18.04+ 上运行，
6  包管理器 (dnf / yum / apt-get) 自动探测，Ubuntu 的包名差异自动映射 (见 `common.sh`)。
7
8  ## 快速开始
9
10 **运行身份**：安装/卸载请用**普通用户**直接运行 (勿加 `sudo`) ——Synopsys Installer
11 拒绝 root 账户 ("The Installer must not be run from a root user account")，root 下
12 安装向引导会直接退出。需要 root 的步骤 (安装 tcsh/依赖库、License 部署的
13  systemd/SELinux 等) 脚本会自动 `sudo` 提权，届时按提示输入密码即可。
14
15 ```bash
16 # 1) 复制配置模板并按需设置安装开关与路径
17 cp install.conf.example install.conf
18 vim install.conf
19
20 # 2) 只安装 VCS 和 Verdi
21 bash install_all.sh --only vcs,verdi
22
23 # 3) 全部安装 (License 部署步会自动 sudo)
24 bash install_all.sh --all
25
26 # 4) 交互式勾选安装
27 bash install_all.sh --interactive
28
29 # 5) 只做 License 部署与校验 (部署需要 root, 直接 sudo 运行)
30 sudo bash license/setup_license.sh
31 bash license/check_license.sh
32
33 # 6) 卸载全部软件 (逐项确认: License 卸载步会自动 sudo)
34 bash uninstall_all.sh
35
36 # 7) 只卸载 VCS 和 Verdi (--force 跳过逐项确认)
37 bash uninstall_all.sh --only vcs,verdi --force
38
39 # 8) 卸载时保留 License 认证
40 bash uninstall_all.sh --skip-license
41
42 # 9) 只安装软件, 不部署/校验 License (License 之后想部署时再单独跑 setup_license.sh)
43 bash install_all.sh --all --skip-license
44
45 # 10) 运行前清理 run.log (默认追加保留历史: 加 --clean-log 则只保留本次运行记录)
46 bash install_all.sh --all --clean-log
47
48 # 11) License 认证完成后, 统一后验证 (文中 3.1~3.10 节; 含 VCS smoke 用例)
49 bash verify_all.sh
50 ```

```

```

52 ## 目录结构
53 ...
54 ...
55 IM_Synopsys_Scripts/install/
56 |__ README.md          本说明
57 |__ CHANGELOG.md       版本记录 (Keep a Changelog 格式)
58 |__ common.sh          公共函数库 (日志 / Installer 命令行自动应答 / bashrc 注入 / 依赖库 / 结果汇总)
59 |__ install.conf.example 顶层配置模板 (复制为 install.conf 后按需修改: install.conf 已被 .gitignore 排除)
60 |__ install_all.sh      顶层入口: 按配置安装 Installer + SCL + 选中的软件 + License 部署
61 |__ uninstall_all.sh    顶层入口: 卸载软件 (--only / --interactive / --force)
62 |__ verify_all.sh       顶层入口: License 认证后统一后验证 (文中 3.x 节)
63
64 |__ tools/
65 |__ |__ install_installer.sh  Installer 5.9
66 |__ |__ install_scl.sh       SCL 2025.03 (License 许可工具)
67 |__ |__ install_vcs.sh       VCS
68 |__ |__ install_verdi.sh     Verdi
69 |__ |__ install_spyglass.sh  Spyglass (含 Docs 文档包自动安装)
70 |__ |__ install_dc.sh        Design Compiler
71 |__ |__ install_formality.sh Formality
72 |__ |__ install_fc.sh        Fusion Compiler
73 |__ |__ install_icc2.sh      ICC2 (含依赖库自动补齐)
74 |__ |__ install_pt.sh        PrimeTime (含 DGCORE_ROOT)
75 |__ |__ install_lc.sh        Library Compiler
76 |__ |__ install_hspice.sh     Hspice
77 |__ |__ uninstall_*.sh       与 install_*.sh 一一对应的卸载脚本 (删 bashrc 块 + 删安装目录)
78
79 |__ license/
80 |__ |__ setup_license.sh     License 认证部署 (拷贝 .dat / LSB / systemd / SELinux)
81 |__ |__ uninstall_license.sh License 认证卸载 (停服务 / 删文件 / 清 SELinux 规则)
82 |__ |__ start_license.sh     License 服务启动脚本 (文中 2.2.1 版参数化)
83 |__ |__ check_license.sh     License 校验 (lstat / 进程 / 特性核对)
84 |__ |__ snps-license.service systemd 服务单元模板 (文中 2.2.2 版参数化)
85
86 |__ utils/
87 |__ |__ diagnose_scl.sh      SCL 安装与 PATH 只读诊断 (不改任何文件)
88 |__ |__ clean_old_config.sh 清理 ~/.bashrc 中旧版 Synopsys 配置 (先备份、可还原)
89
90 |__ verify/
91 |__ |__ vcs_smoke/           VCS 启动验证工程 (run_smoke.sh + 示例设计, 产物拆分 work/ 与 result/)
92
93 ...

```



```

90  ## 重要说明 (务必阅读)
91
92  1. **实测声明**: 核心安装/卸载链路已在 **CentOS 7** 实测验证 ('install_all.sh' --skip-license --clean-log、
93  'uninstall_all.sh'); AlmaLinux 8 / Ubuntu 18.04+ 为跨发行版适配目标, 包管理解与包名差异已自动探测
94  (见 'common.sh'), 建议先在虚拟机/测试机完整跑一遍再用于正式部署。软件安装采用 Installer 5.9
95  **命令行模式** ('-source' / '-target' 直传, Site ID 与交互应答由 stdin 管道喂入, 无交互问答),
96  安装异常时查看 'run.log' 最新 '[SESSION]' 段。
97  2. **无 expect 依赖**: 软件安装与 Installer 自身安装均不依赖 expect, 全部通过 stdin 管道自动应答。
98  3. **发行版差异**: 除包管理外, 仅 SELinux 一段有差异—Ubuntu 默认无 SELinux
99  (AppArmor), 'setup_license.sh' 检测到非 Enforcing 会自动跳过规则配置, 无需改动;
100  CentOS 7 的 SELinux 行为与 AlmaLinux 一致。若在 Ubuntu 24.04 安装 ICC2 依赖
101  'libns1' 失败, 请将 'common.sh' 中 'apt-get:libns1 -> libns11' 改为 'libns12';
102  32 位系统如遇 lmgrd 缺库, 需另装 'libc6-1386' (Ubuntu) / 'glibc.1686' (RHEL 系)。
103  4. **路径**: 'PKG_BASE' 为安装包存放目录 (文中示例)
104  '/home/<user>/下载/AlmaLinux_ASIC/Synopsys/', 请按你的实际情况修改;
105  'INSTALL_ROOT' 默认 '/home/<user>/usr/synopsys' (与 'install.conf.example' 一致; 文中示例为
106  '/mnt/asic/Software/Synopsys', 按你的实际环境修改)。
107  5. **License 文件**: (Synopsys.dat)需通过官方正规渠道或公司授权获得, 脚本不会
108  生成或下载 License 文件: 部署前请放入本脚本目录下的 'license' 文件夹
109  ('install/license/Synopsys.dat'), 运行 'setup_license.sh' 时脚本会自动复制到
110  '$INSTALL_ROOT/scl/$SCL_VERSION/admin/license/Synopsys.dat' 并部署启动。
111  **注意**: 'license/Synopsys.dat' 为本地授权文件, 已被 '.gitignore' 排除, 请勿提交到仓库。
112  6. **主机名一致性**: 'HOSTNAME'、'SNPSLMD_LICENSE_FILE' 端口@主机名、License 文件
113  中登记的 SERVER 主机名三者必须一致 (文中 2.1 节), 修改主机名后需重启或
114  'hostnamectl set-hostname' 后重新登录。
115  7. **bashrc 注入**: 每个软件脚本只注入自己的环境变量段 (带 '# >>> SYNOPSYS:xxx'
116  标记, 等等, 重复执行不会叠加; 内容一致时静默跳过)。环境变量写入**实际登录用户**
117  (sudo 发起人, 即 'SUDO_USER') 的 '~/.bashrc', 而非 root 的 '/root/.bashrc';
118  安装/卸载完成后脚本会自动 'source' 该 bashrc, 无需手动执行。安装/卸载前还会
119  **自动检测并清理 bashrc 中的旧版 Synopsys 配置** (如 2018 旧版块, 先备份再注释,
120  防止旧 PATH 干扰新版本)。SCL 的 PATH 不带 '/bin' (文中 1.3 差异点 2)、
121  SPYGLASS_HOME 与 HSPICE_HOME 路径结构特殊 (差异点 3、5), 脚本已按文中处理。
122  8. **内存**: Synopsys 软件编译/跑大型设计非常吃内存 (文中 3.4、总结), 请按需配备。
123  9. **卸载安全**: 卸载脚本默认逐项确认 ('--force' 可跳过); 每个软件卸载 = 移除 bashrc
124  对应环境变量块 + 删除 '$INSTALL_ROOT' 下安装目录 (含附加目录与 Installer 状态目录
125  'installer' 的彻底清理)。License 文件 (Synopsys.dat) 为官方渠道获得的资产,
126  'uninstall_license.sh' 删除前会单独确认, 建议先备份; SCL 卸载前会自动停止
127  snps-license 服务。 '--skip-license' 可保留 License 认证。
128  10. **结果汇总与中断恢复**: 安装/卸载完成 (或被 Ctrl+C 中断) 时, 脚本都会输出结果汇总
129  一每款软件一行 (状态 / 空间占用 / 安装用时), 末尾统计**总空间占用**与**总用时**:
130  Installer 由预装阶段处理, 汇总中显示“已预装/已就绪”, '--skip-license' 时汇总标题为
131  「结果汇总 (已安装未认证)」。安装过程可随时 Ctrl+C 中断, 已完成的软件不会重复安装
132  (需等待跳过), 重新运行 'install_all.sh' 即从未完成处继续。
133  11. **启动验证**: 'verify_all.sh' 在 License 认证完成后统一执行文中 3.x 节各软件启动验证:
134  其中 VCS 使用内置 smoke 工程 ('verify/vcs_smoke', 编译 + 仿真生成 FSDB 波形,
135  产物拆分 'work/' 过程文件与 'result/' 最终结果)。依赖库 (tcsh 等) 由 'pkg_install'
136  自动安装 (RHEL 系先 'rpm -q' 检查、已装即跳过, 仅装缺失包; 函数内部自动 sudo 提权)。
137  12. **运行日志**: 所有脚本的 '[INFO]' / '[WARN]' / '[ERROR]' / '[OK]' 输出统一保存到
138  本目录下的 **run.log** (追加写入、保留历史, 每次运行带 '[SESSION]' 分隔头;
139  Installer 的原始输出与报错也会实时写入 'run.log', 安装异常时从最新
140  '[SESSION]' 段后直接查看向寻原始信息), 如需清空旧日志, 运行
141  'install_all.sh' / 'uninstall_all.sh' 时加 '--clean-log' 参数; 也可用环境变量
142  'LOG_PATH' 指定其他日志位置。Installer 5.9 强制生成的 'installer.log' 内容与
143  'run.log' 重复, 脚本运行后会**自动删除**, 不会残留多余文件。License 服务的
144  运行日志仍为 '$INSTALL_ROOT/scl/$SCL_VERSION/admin/logs/snps_license.log'
145  (文中 2.2.1 节)。
146
147  ## 许可
148
149  MIT License (见上级目录 'LICENSE')。脚本仅提供自动化安装与运维能力,
150  **不包含、不生成任何破解、License 密钥或授权内容**: (Synopsys.dat)需通过官方
151  正规渠道或公司授权获得 (本地文件已被 '.gitignore' 排除, 请勿提交)。
152

```

除需 root 权限的步骤 (依赖库 / License 部署等) 脚本在自动 sudo 提权时需用户按提示输入密码之外, 无需其他人工干预。建议运行脚本前执行“**sudo -v**”指令。

运行该脚本前须先将《install.conf.example》改名为《install.conf》, 并根据自身情况设置用户名、路径、软件安装包版本、License 文件特性等配置信息。

```

install.conf  install.conf.example
1  #!/bin/bash
2  #
3  # install.conf.example - 顶层配置模板 (对应文中 1.1~1.14 节安装内容)
4  # 依据《【老刘炼芯】十款Synopsys芯片设计软件的部署》
5  #
6  # 使用方式：复制本文件为 install.conf 后按需修改
7  # cp install.conf.example install.conf && vim install.conf
8  # (install.conf 为本地配置，已被 .gitignore 排除，不会提交到仓库)
9  #
10
11 # ----- 安装开关 (yes / no) -----
12 # Installer (安装引导工具，必须先装)
13 INSTALL_INSTALLER=yes
14 # SCL (License 许可工具，建议在装其他软件前先完成 License 认证)
15 INSTALL_SCL=yes
16 # 十款芯片设计软件
17 INSTALL_VCS=yes
18 INSTALL_VERDI=yes
19 INSTALL_SPYGLASS=yes
20 INSTALL_DC=yes
21 INSTALL_FORMALITY=yes
22 INSTALL_FC=yes
23 INSTALL_ICC2=yes
24 INSTALL_PT=yes
25 INSTALL_LC=yes
26 INSTALL_HSPICE=yes
27 # 完成 SCL 安装后是否立即执行 License 认证部署 (如License已确认准备好则推荐 yes，见文中 1.4 节末建议，否则建议为 no)
28 SETUP_LICENSE_AFTER_SCL=no
29
30 # ----- 路径 -----
31 # 安装包存放目录，请根据您的具体情况修改 (文中：/home/<user>/下载/AlmaLinux_ASIC/Synopsys/)
32 PKG_BASE="/home/<your user>/Downloads/Synopsys"
33 # 安装目标根目录，请根据您的具体情况修改 (文中 1.3 节 Synopsys_DIR)
34 INSTALL_ROOT="/home/<your user>/usr/synopsys"
35 # 脚本自身所在目录 (自动探测，无需修改)
36 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
37
38 # ----- 版本 (文中 1.1 / 表一 / 1.3 节) -----
39 INSTALLER_VERSION="5.9"
40 SCL_VERSION="2025.03"
41 # SCL 安装包目录名后缀 (按实际目录名填写；如目录为 vcs_vV-2023.12-SP1 则填 "V-2023.12-SP1"，无后缀留空)
42 SCL_PKG_SUFFIX=""
43 VCS_VERSION="V-2023.12-SP1"
44 VERDI_VERSION="V-2023.12-SP2"
45 SPYGLASS_VERSION="V-2023.12-SP1"
46 DC_VERSION="V-2023.12-SP3"
47 FM_VERSION="V-2023.12-SP3"
48 FC_VERSION="V-2023.12"
49 ICC2_VERSION="V-2023.12-SP3"
50 PT_VERSION="V-2023.12-SP5-1"
51 LC_VERSION="V-2023.12-SP3"
52 HSPICE_VERSION="V-2023.12-SP2"
53 # VIP 版本 (文中 1.3 节 VIP 段：DESIGNWARE_HOME=$Synopsys_DIR/vip/V-2023.12，无 SP 后缀)
54 VIP_VERSION="V-2023.12"
55
56 # ----- License 相关 (文中 1.3 / 2.1 / 2.2 节) -----
57 # 主机名：须与本机 hostname 及 License 文件中登记的 SERVER 主机名一致 (如 1m-nuc17)
58 HOSTNAME=""
59 # License 服务端口 (文中 30000@主机名)
60 LICENSE_PORT="30000"
61 # Site ID (文中 1.2 通用流程第 5 步，默认 000)
62 # Site ID number (Installer 安装时按此值使用：Synopsys 默认 000，
63 # 如你的 License 证书右上角 Site ID 不是 000，请改为实际值)
64 SITE_ID="000"
65 # 当前登录用户 (systemd 服务 User/Group 用；root 执行时取 sudo 发起人，一般无需修改)
66 RUN_USER="${SUDO_USER:-$(logname 2>/dev/null || echo root)}"
67 # License 文件 (需通过官方正规渠道或公司授权获得；部署前请放入脚本目录下的 license 文件夹，
68 # 即 $SCRIPT_DIR/license/Synopsys.dat；运行 setup_license.sh 时脚本会自动复制到下面的部署路径)
69 LICENSE_DAT_SOURCE="$SCRIPT_DIR/license/Synopsys.dat"
70 # 部署后的 License 文件位置 (自动拼接，一般无需修改)
71 LICENSE_FILE="$INSTALL_ROOT/scl/$SCL_VERSION/admin/license/Synopsys.dat"
72 LOG_DIR="$INSTALL_ROOT/scl/$SCL_VERSION/admin/logs"
73 LOG_FILE="$LOG_DIR/snps_license.log"
74
75 # ----- Installer 安装方式 -----
76 # 软件安装一律使用 Installer 5.9 命令行模式 (-source/-target 直传，无交互)，
77 # Site ID / 附加交互应答通过 stdin 管道喂入 (见 common.sh install_one)，不依赖 expect。
78
79 # ----- 可选：VIP 环境变量 (文中 1.3 节 bashrc 有 VIP 段，但本篇未安装 VIP 软件包) -----
80 # 若你另外安装了 VIP 包，可置 yes 注入 DESIGNWARE_HOME
81 ENABLE_VIP_ENV=yes
82
83 # ----- 校验特性清单 (check_license.sh 用；按需增删) -----
84 # 格式：软件名:特性名 (多个特性用空格分隔)
85 CHECK_FEATURES="
86 VCS:VCSCompiler_Net VCSRuntime_Net VCS-BASE-COMPILE VCS-BASE-RUNTIME
87 Verdi:Verdi VERDI-ULTRA
88 Spyglass:OEM-spyglass_Lc
89 DC:Design-Compiler
90 Formality:Formality Formality-Advanced
91 FC:Fusion-Compiler-AG Fusion-Compiler-BE Fusion-Compiler-FE Fusion-Compiler-ML
92 ICC2:ICCompilerII ICCompilerIIs-AdvFeatures ICCompilerIIs-AdvFeatures
93 PrimeTime:PrimeTime PrimeTime-Plus
94 LC:Library-Compiler Library-Compiler-Advanced
95 Hspice:hspice
96 "

```

目前老刘已在 CentOS Linux release 7.9.2009 系统中完成测试的指令效果如下：

(1) 安装部署全部 Synopsys 软件且不立刻部署校验 License，日志文件中只保留本次运行内容，过往日志信息予以清空。安装过程中可随时“**Ctrl + C**”中止任务，再次重复此指令时会自动跳过已安装软件，继续执行未完成的安装过程：

bash install_all.sh --all --skip-license --clean-log

```

1
2
3 [SESSION] 2026-10-01 17:20:49 命令: install_all.sh
4
5
6 Synopsys 软件自动化安装
7 安装根目录: /home/liumin/usr/synopsys
8 安装包目录: /home/liumin/Downloads/Synopsys
9 本次计划: installer scl vcs verdi spyglass dc formality fc icc2 pt lc hspice
10 提示: License 部署与依赖库步骤需要 root 权限 (自动 sudo)。
11 如遇 sudo 等待密码超时，请先执行 sudo -V 输入一次密码 (默认 15 分钟内有效)。
12
13 [INFO] 2026-10-01 17:20:49 已按 ENABLE_VIP_ENV=yes 写入 # >>> SYNOPSISVIP 环境变量块 (DESIGNWARE_HOME=/home/liumin/usr/synopsys/vip/V-2023.12)
14
15 >>> Installer 未就绪，先自动安装 (install_installer.sh) ...
16 [OK] 2026-10-01 17:20:49 已向 /home/liumin/.bashrc 写入 SYNOPSIS:INSTALLER 环境变量块
17 [INFO] 2026-10-01 17:20:49 Installer 安装: /home/liumin/Downloads/Synopsys/Installer_v5.9/SynopsysInstaller_v5.9.run -> /home/liumin/usr/synopsys/installer
18
19 *****
20 #
21 # Synopsys Installer Self-Extracting Executable
22 #
23 # This script extracts and installs Synopsys Installer 5.9 into
24 # the given directory
25 #
26 # For help type:
27 #
28 # SynopsysInstaller_v5.9.run -help
29 #
30 *****
31
32 Please specify installation directory [:] Installing Synopsys Installer 5.9 into the directory '/home/liumin/usr/synopsys/installer'...
33
34 Creating installation directory '/home/liumin/usr/synopsys/installer'...
35 Unpacking: SynopsysInstaller.tgz ...
36
37 Installation complete.
38
39 [OK] 2026-10-01 17:20:50 Installer 安装完成: /home/liumin/usr/synopsys/installer/installer
40 [INFO] 2026-10-01 17:20:50 Installer 安装完成，启动验证见文中 3.x 节 (verify_all.sh 统一执行)。
41
42 >>> 安装 scl (install_scl.sh) ...
43 [OK] 2026-10-01 17:20:50 已向 /home/liumin/.bashrc 写入 SYNOPSIS:SCL 环境变量块
44 [INFO] 2026-10-01 17:20:50 开始安装 scl: 源目录 /home/liumin/Downloads/Synopsys/scl_v2025.03
45
46 Synopsys (R) Installer
47 Version 5.9.0
48
49 Copyright (C) 2005 - 2026 Synopsys, Inc.
50 This software and the associated documentation are proprietary to Synopsys, Inc.
51 This software may only be used in accordance with the terms and conditions of a
52 written license agreement with Synopsys, Inc. All other use, reproduction, or
53 distribution of this software is strictly prohibited.
54
55 Extracting release information from scl_v2025.03_linux64.spf
56 Extracting release information from scl_v2025.03_linux64.spf successfully
57 Extracting release information from scl_v2025.03_common.spf
58 Extracting release information from scl_v2025.03_common.spf successfully
59
60 Your site ID number is in the upper-right corner of your Synopsys license
61 key certificate. If you have trouble locating it, contact your Synopsys
62 representative.
63
64 Site ID number [000]: *****
65 ***** Installation Summary *****
66 Product: SCL (Synopsys Common Licensing) (scl)
67 Release: 2025.03
68 Platforms: linux64 common
69 Disk Space: 105 MB
70 Target Dir: /home/liumin/usr/synopsys/scl/2025.03
71
72 Licensed Products communicate with Synopsys servers for the purpose of providing software updates,
73 detecting software piracy and verifying that customers are using Licensed Products in conformity with
74 the applicable License Key for such Licensed Products. Synopsys will use information gathered in
75 connection with this process to deliver software updates and pursue software pirates and infringers.
76 *****
77 If the information is correct, continue with the installation.
78 Accept, Install? [yes]: INFO: Prepare for installing 2025.03 release of product scl ...
79 INFO: Extracting linux64 package from scl_v2025.03_linux64.spf ...
80 INFO: Extracting common package from scl_v2025.03_common.spf ...
81
82
83
84
85
86
87
88
89 >>> 已使用 --skip-license，跳过 License 部署与统一启动验证 (如需验证，请先部署 License 后运行 bash verify_all.sh)
90
91
92 结果汇总 (已安装未认证)
93
94 软件 状态 空间占用 用时
95 -----
96 installer 已安装 42M 1s
97 scl 已安装 101M 3s
98 vcs 已安装 14G 4m 43s
99 verdi 已安装 23G 8m 03s
100 spyglass 已安装 3.4G 13m 22s
101 dc 已安装 12G 3m 49s
102 formality 已安装 3.5G 1m 05s
103 fc 已安装 14G 4m 27s
104 icc2 已安装 9.7G 3m 19s
105 pt 已安装 7.2G 2m 21s
106 lc 已安装 3.2G 1m 03s
107 hspice 已安装 5.8G 1m 47s
108
109 -----
110 总空间占用 : 95.9G
111 总用时 : 44m 04s
112
113 下一步:
114 1) bash license/check_license.sh # License 校验
115

```

从上图可以看出，在老刘这台测试电脑上，十款 Synopsys 芯片设计软件利用 Bash (Shell) 脚本不间断地执行自动化安装，仍需耗时长达 **44 分钟**，如果由人手动操作，则耗时更长耗力更巨。十款 Synopsys 软件安装后空间占用 **95.9GB**，应注意预留充足的硬盘空间。

(2) 卸载所有 Synopsys 芯片设计软件及其安装认证组件。该指令默认在每删除一个组件时都会交互式询问用户确认操作，如不希望逐项确认，可添加 **--force** 条件。

bash uninstall_all.sh

```
run.log
1
2=====
3[SESSION] 2026-10-01 18:40:33 命令: uninstall_all.sh
4=====
5=====
6Synopsys 软件卸载
7安装根目录: /home/liumin/usr/synopsys
8本次计划: hspice lc pt icc2 fc formality dc spyglass verdi vcs scl installer
9注意: bashrc 中对应环境变量块将一并移除; License 认证在本次计划内
10=====
11
12>>> 卸载 hspice (uninstall_hspice.sh) ...
13[ OK ] 2026-10-01 18:40:33 已从 /home/liumin/.bashrc 移除 SYNOPSYS:HSPICE 块
14将删除 Hspice: /home/liumin/usr/synopsys/hspice (占用 5.8G)
15确认删除? [y/N] [ OK ] 2026-10-01 18:40:37 已删除 Hspice: /home/liumin/usr/synopsys/hspice
16[ OK ] 2026-10-01 18:40:37 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/hspice
17[INFO] 2026-10-01 18:40:37 Hspice 卸载完成。
18[INFO] 2026-10-01 18:40:37 => hspice 卸载用时: 4s
19
20>>> 卸载 lc (uninstall_lc.sh) ...
21[ OK ] 2026-10-01 18:40:38 已从 /home/liumin/.bashrc 移除 SYNOPSYS:LC 块
22将删除 Library Compiler: /home/liumin/usr/synopsys/lc (占用 3.2G)
23确认删除? [y/N] [ OK ] 2026-10-01 18:40:42 已删除 Library Compiler: /home/liumin/usr/synopsys/lc
24[ OK ] 2026-10-01 18:40:42 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/lc
25[INFO] 2026-10-01 18:40:42 LC 卸载完成。
26[INFO] 2026-10-01 18:40:42 => lc 卸载用时: 5s
27
28>>> 卸载 pt (uninstall_pt.sh) ...
29[ OK ] 2026-10-01 18:40:42 已从 /home/liumin/.bashrc 移除 SYNOPSYS:PRIMETIME 块
30将删除 PrimeTime: /home/liumin/usr/synopsys/prime (占用 7.2G)
31确认删除? [y/N] [ OK ] 2026-10-01 18:40:54 已删除 PrimeTime: /home/liumin/usr/synopsys/prime
32[ OK ] 2026-10-01 18:40:54 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/prime
33[INFO] 2026-10-01 18:40:54 PrimeTime 卸载完成。
34[INFO] 2026-10-01 18:40:54 => pt 卸载用时: 12s
35
36>>> 卸载 icc2 (uninstall_icc2.sh) ...
37[ OK ] 2026-10-01 18:40:54 已从 /home/liumin/.bashrc 移除 SYNOPSYS:ICC2 块
38将删除 ICC2: /home/liumin/usr/synopsys/icc2 (占用 9.7G)
39确认删除? [y/N] [ OK ] 2026-10-01 18:41:10 已删除 ICC2: /home/liumin/usr/synopsys/icc2
40[ OK ] 2026-10-01 18:41:10 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/icc2
41[INFO] 2026-10-01 18:41:10 ICC2 卸载完成。
42[INFO] 2026-10-01 18:41:10 => icc2 卸载用时: 16s
43
44>>> 卸载 fc (uninstall_fc.sh) ...
45[ OK ] 2026-10-01 18:41:10 已从 /home/liumin/.bashrc 移除 SYNOPSYS:FC 块
46将删除 Fusion Compiler: /home/liumin/usr/synopsys/fusioncompiler (占用 14G)
47确认删除? [y/N] [ OK ] 2026-10-01 18:41:37 已删除 Fusion Compiler: /home/liumin/usr/synopsys/fusioncompiler
48[ OK ] 2026-10-01 18:41:37 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/fusioncompiler
49[INFO] 2026-10-01 18:41:37 FC 卸载完成。
50[INFO] 2026-10-01 18:41:37 => fc 卸载用时: 27s
51
52>>> 卸载 formality (uninstall_formality.sh) ...
53[ OK ] 2026-10-01 18:41:37 已从 /home/liumin/.bashrc 移除 SYNOPSYS:FORMALITY 块
54将删除 Formality: /home/liumin/usr/synopsys/fm (占用 3.5G)
55确认删除? [y/N] [ OK ] 2026-10-01 18:41:50 已删除 Formality: /home/liumin/usr/synopsys/fm
56[ OK ] 2026-10-01 18:41:50 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/fm
57[INFO] 2026-10-01 18:41:50 Formality 卸载完成。
58[INFO] 2026-10-01 18:41:50 => formality 卸载用时: 13s
59
60>>> 卸载 dc (uninstall_dc.sh) ...
61[ OK ] 2026-10-01 18:41:50 已从 /home/liumin/.bashrc 移除 SYNOPSYS:DC 块
62将删除 Design Compiler: /home/liumin/usr/synopsys/syn (占用 12G)
63确认删除? [y/N] [ OK ] 2026-10-01 18:41:57 已删除 Design Compiler: /home/liumin/usr/synopsys/syn
64[ OK ] 2026-10-01 18:41:57 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/syn
65[INFO] 2026-10-01 18:41:57 DC 卸载完成。
66[INFO] 2026-10-01 18:41:57 => dc 卸载用时: 7s
67
68>>> 卸载 spyglass (uninstall_spyglass.sh) ...
69[ OK ] 2026-10-01 18:41:57 已从 /home/liumin/.bashrc 移除 SYNOPSYS:SPYGLASS 块
70将删除 Spyglass: /home/liumin/usr/synopsys/spyglass (占用 3.4G)
71确认删除? [y/N] [ OK ] 2026-10-01 18:41:59 已删除 Spyglass: /home/liumin/usr/synopsys/spyglass
72[ OK ] 2026-10-01 18:41:59 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/spyglass
73将删除 Spyglass 附加包 ufe_optional_spyglass-vcs: /home/liumin/usr/synopsys/ufe_optional_spyglass-vcs (占用 36G)
74确认删除? [y/N] [ OK ] 2026-10-01 18:42:10 已删除 Spyglass 附加包 ufe_optional_spyglass-vcs: /home/liumin/usr/synopsys/ufe_optional_spyglass-vcs
75[ OK ] 2026-10-01 18:42:10 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/ufe_optional_spyglass-vcs
76[INFO] 2026-10-01 18:42:10 Spyglass 卸载完成。
77[INFO] 2026-10-01 18:42:10 => spyglass 卸载用时: 13s
78
```

```
run.log
79 >>> 卸载 verdi (uninstall_verdi.sh) ...
80 [ OK ] 2026-10-01 18:42:10 已从 /home/liumin/.bashrc 移除 SYNOPSIS:VERDI 块
81 将删除 Verdi: /home/liumin/usr/synopsys/verdi (占用 23G)
82 确认删除? [y/N] [ OK ] 2026-10-01 18:42:14 已删除 Verdi: /home/liumin/usr/synopsys/verdi
83 [ OK ] 2026-10-01 18:42:14 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/verdi
84 将删除 Verdi 附加目录 verdi_supp: /home/liumin/usr/synopsys/verdi_supp (占用 8.0K)
85 确认删除? [y/N] [ OK ] 2026-10-01 18:42:16 已删除 Verdi 附加目录 verdi_supp: /home/liumin/usr/synopsys/verdi_supp
86 [ OK ] 2026-10-01 18:42:16 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/verdi_supp
87 [INFO] 2026-10-01 18:42:16 Verdi 卸载完成
88 [INFO] 2026-10-01 18:42:16 => verdi 卸载用时: 6s
89
90 >>> 卸载 vcs (uninstall_vcs.sh) ...
91 [ OK ] 2026-10-01 18:42:16 已从 /home/liumin/.bashrc 移除 SYNOPSIS:VCS 块
92 将删除 VCS: /home/liumin/usr/synopsys/vcs (占用 14G)
93 确认删除? [y/N] [ OK ] 2026-10-01 18:42:20 已删除 VCS: /home/liumin/usr/synopsys/vcs
94 [ OK ] 2026-10-01 18:42:20 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/vcs
95 [WARN] 2026-10-01 18:42:20 VCS DVE 运行日志目录 DVEfiles 未安装 (目录不存在: /home/liumin/usr/synopsys/DVEfiles), 跳过
96 将删除 VCS Docs 安装状态记录: /home/liumin/usr/synopsys/.installer/vcs_docs (占用 8.0K)
97 确认删除? [y/N] [ OK ] 2026-10-01 18:42:22 已删除 VCS Docs 安装状态记录: /home/liumin/usr/synopsys/.installer/vcs_docs
98 [INFO] 2026-10-01 18:42:22 VCS 卸载完成
99 [INFO] 2026-10-01 18:42:22 => vcs 卸载用时: 6s
100
101 >>> 卸载 scl (uninstall_scl.sh) ...
102 [ OK ] 2026-10-01 18:42:22 已从 /home/liumin/.bashrc 移除 SYNOPSIS:SCL 块
103 将删除 SCL: /home/liumin/usr/synopsys/scl (占用 101M)
104 确认删除? [y/N] [ OK ] 2026-10-01 18:42:27 已删除 SCL: /home/liumin/usr/synopsys/scl
105 [ OK ] 2026-10-01 18:42:27 已清理 Installer 安装状态: /home/liumin/usr/synopsys/.installer/scl
106 [ OK ] 2026-10-01 18:42:27 已清理空 Installer 状态目录: /home/liumin/usr/synopsys/.installer
107 [INFO] 2026-10-01 18:42:27 SCL 卸载完成
108 [INFO] 2026-10-01 18:42:27 => scl 卸载用时: 5s
109
110 >>> 卸载 installer (uninstall_installer.sh) ...
111 [ OK ] 2026-10-01 18:42:27 已从 /home/liumin/.bashrc 移除 SYNOPSIS:INSTALLER 块
112 将删除 Installer: /home/liumin/usr/synopsys/installer (占用 42M)
113 确认删除? [y/N] [ OK ] 2026-10-01 18:42:29 已删除 Installer: /home/liumin/usr/synopsys/installer
114 [INFO] 2026-10-01 18:42:29 Installer 卸载完成
115 [INFO] 2026-10-01 18:42:29 => installer 卸载用时: 2s
116
117 >>> 卸载 License 认证 (license/uninstall_license.sh, 需要 root, 将自动 sudo 提权, 请按提示输入密码) ...
118 == [1/5] 停止并禁用 systemd 服务 ==
119 [ OK ] 2026-10-01 18:42:34 已停止并移除 /etc/systemd/system/snps-license.service
120 == [2/5] 删除 start_license.sh ==
121 [ OK ] 2026-10-01 18:42:34 已删除 /home/liumin/usr/synopsys/scl/2025.03/admin/license/start_license.sh
122 == [3/5] 删除 License 文件 (Synopsys.dat) ==
123 [WARN] 2026-10-01 18:42:34 License 文件不存在: /home/liumin/usr/synopsys/scl/2025.03/admin/license/Synopsys.dat, 跳过
124 == [4/5] 清理 SELinux 自定义规则 (仅 Enforcing 时) ==
125 [ OK ] 2026-10-01 18:42:35 已移除 SELinux 自定义规则 (usr_t / bin_t)
126 == [5/5] 检查 lmgrd 残留进程 ==
127 [ OK ] 2026-10-01 18:42:35 无 lmgrd 进程残留
128 [INFO] 2026-10-01 18:42:35 License 认证已卸载. bashrc 中 SNPSLMD_LICENSE_FILE / Synopsys_DIR 随 SCL 块移除 (见 uninstall_scl.sh).
129
130 >>> 先底清理附加目录 (ufe_optional_spyglass-vcs / verdi_supp / DVEfiles) ...
131 uninstall_all.sh: line 149: local: can only be used in a function
132 [WARN] 2026-10-01 18:42:35 Installer 状态目录 .installer 未安装 (目录不存在: /home/liumin/usr/synopsys/.installer), 跳过
133 [ OK ] 2026-10-01 18:42:36 已自动重新加载 /home/liumin/.bashrc
134
135 =====
136 结果汇总 (卸载)
137 =====
138 软件      状态      空间占用 用时
139 -----
140 hspIcse   已卸载    -         4s
141 lc         已卸载    -         5s
142 pt        已卸载    -        12s
143 icc2      已卸载    -        16s
144 fc        已卸载    -        27s
145 formality 已卸载    -        13s
146 dc        已卸载    -         7s
147 spyglass  已卸载    -        13s
148 verdi     已卸载    -         6s
149 vcs       已卸载    -         6s
150 scl       已卸载    -         5s
151 installer 已卸载    -         2s
152 -----
153 总用时 : 2m 03s
154
155 后续建议:
156 1) env | grep -i synopsys # 确认环境变量已清除 (或重新登录)
157 2) 如需同时清理安装包目录: rm -rf /home/liumin/Downloads/Synopsys (请确认不再需要)
158
```

(3) License 一键部署与校验:

sudo bash license/setup_license.sh


```
run.log
1440 =====
1441 [SESSION] 2026-10-02 14:26:00 命令: license/setup_license.sh
1442 =====
1443 == [1/7] 部署 start_license.sh ==
1444 [ OK ] 2026-10-02 14:26:00 hostname = lm-nuc17, 一致
1445 == [2/7] 安装 LSB 兼容层 (redhat-lsb-core / lsb-release) ==
1446 [INFO ] 2026-10-02 14:26:00 安装依赖 (yum): redhat-lsb-core
1447 [INFO ] 2026-10-02 14:26:00 依赖已全部安装 (redhat-lsb-core), 跳过安装
1448 == [3/7] 部署 License 文件 ==
1449 [ OK ] 2026-10-02 14:26:00 已部署 License 文件: /home/liumin/usr/synopsys/scl/2025.03/admin/license/Synopsys.dat (请通过官方正规渠道或公司授权获得)
1450 == [4/7] 部署 start_license.sh ==
1451 [ OK ] 2026-10-02 14:26:00 已生成: /home/liumin/usr/synopsys/scl/2025.03/admin/license/start_license.sh
1452 == [5/7] 创建 systemd 服务 snps-license.service ==
1453 [ OK ] 2026-10-02 14:26:00 已创建并启用: /etc/systemd/system/snps-license.service (开机 10 秒网络延时、30 秒重启动时, 见文中 2.2.2 节可自行修订)
1454 == [6/7] SELinux 上下文处理 ==
1455 [ OK ] 2026-10-02 14:26:00 SELinux 处于 Enforcing, 按文中 2.2.3 节添加 usr_t / bin_t 规则
1456 检查 lmgrd 上下文 (应显示 bin_t):
1457 -rw-r-xr-x 1 liumin liumin unconfined_u:object_r:bin_t:s0 /home/liumin/usr/synopsys/scl/2025.03/linux64/bin/lmgrd
1458 == [7/7] 启动 License 服务校验 ==
1459 ● snps-license.service - Synopsys License Server
1460 Loaded: loaded (/etc/systemd/system/snps-license.service; enabled; vendor preset: disabled)
1461 Active: active (running) since 五 2026-10-02 14:26:22 CST; 3s ago
1462 Process: 6826 ExecStartPre=/home/liumin/usr/synopsys/scl/2025.03/admin/license/start_license.sh start (code=exited, status=0/SUCCESS)
1463 Process: 6810 ExecStartPre=/bin/sleep 10 (code=exited, status=0/SUCCESS)
1464 Main PID: 6831 (lmgrd)
1465 Tasks: 2
1466 CGroup: /system.slice/snps-license.service
1467 =====
1468 [1/4] License 服务进程检查
1469 =====
1470 ✓ lmgrd 进程运行中 (PID: 6831)
1471 =====
1472 [2/4] 端口监听检查 (30000)
1473 =====
1474 ✓ 端口 30000 正在监听
1475 =====
1476 [3/4] lmstat 服务状态检查 (文中 2.1 节)
1477 =====
1478 License server status: 30000@lm-nuc17
1479 lm-nuc17: license server UP (MASTER) v11.19.7
1480 ✓ lmstat 输出正常 (License Server UP)
1481 =====
1482 [4/4] 软件特性核对 (文中 3.5 节, grep License 文件排查缺失特性)
1483 说明: 特性缺失仅影响对应新工艺/附加功能, 工具主体仍可运行;
1484 例: Formality-NewTech2 缺失不影响 Formality 主体启动 (见文中 3.5 节)
1485 =====
1486 ✓ VCS: 特性 VCSCompiler.Net 存在
1487 ✓ VCS: 特性 VCSRuntime.Net 存在
1488 ✓ VCS: 特性 VCS-BASE-COMPILER 存在
1489 ✓ VCS: 特性 VCS-BASE-RUNTIME 存在
1490 ✓ Verdi: 特性 Verdi 存在
1491 ✓ Verdi: 特性 VERDI-ULTRA 存在
1492 ✓ Spyglass: 特性 oemunlock 存在
1493 ✓ DC: 特性 Design-Compiler 存在
1494 ✓ Formality: 特性 Formality 存在
1495 ✓ Formality: 特性 Formality-Advanced 存在
1496 ✓ FC: 特性 Fusion-Compiler-AG 存在
1497 ✓ FC: 特性 Fusion-Compiler-BE 存在
1498 ✓ FC: 特性 Fusion-Compiler-FE 存在
1499 ✓ FC: 特性 Fusion-Compiler-ML 存在
1500 ✓ IC2: 特性 ICCompilerII 存在
1501 ✓ IC2: 特性 ICCompilerIIs-AdvFeatures 存在
1502 ✓ IC2: 特性 ICCompilerIIs-AdvFeatures 存在
1503 ✓ PrimeTime: 特性 PrimeTime 存在
1504 ✓ PrimeTime: 特性 PrimeTime-Plus 存在
1505 ✓ LC: 特性 Library-Compiler 存在
1506 ✓ LC: 特性 Library-Compiler-Advanced 存在
1507 ✓ Hspice: 特性 hspice 存在
1508 =====
1509 校验完成: PASS=3 FAIL=0
1510 日志文件: /home/liumin/usr/synopsys/scl/2025.03/admin/logs/snps_license.log (可打开检查确认, 见文中 2.1 节)
1511 =====
1512
1513 [ OK ] 2026-10-02 14:27:04 License 认证部署完成, 命令对照 (文中 2.2 节):
1514 systemctl status snps-license.service # 查看服务状态
1515 sudo systemctl restart snps-license # 重启服务
1516 start_license.sh {start|stop|restart|status} # 或直接使用脚本 (alias snps-license)
1517
```

示例脚本工程已上传 Github, 放在《LM_Synopsys_Scripts》仓库的《install》目录下:

https://github.com/patton016/LM_Synopsys_Scripts

不方便上 GitHub 的话, 自动化部署脚本也可以从此地址下载:

https://www.fpga.pw/download/LM_Synopsys_Scripts_Install.rar

您也可以关注本公众号回复“Synopsys Install”或“Synopsys 安装”获得压缩包下载链接。

由于条件所限, 老刘目前仅在 CentOS Linux release 7.9.2009 系统中对该脚本予以了自动化运行测试确认 PASS。当前脚本主要用于满足老刘自身的软件部署需求, 如逢有兴趣的朋友愿意共同参与优化维护, 甚为欢迎!

【总结】

本篇笔记示例了十款 Synopsys 芯片设计软件及其安装引导、License 工具在 AlmaLinux 8.10 系统中的部署过程, 其中除 SCL 选择 2025.03 版本以外, Installer 版本选择为 5.9, 其余软件包版本均为 2023.12。

Synopsys 各软件均可通过 Synopsys Installer 安装, 遵循模板化的通用安装流程, 仅部分细节和环境依赖的处理略有区别, 执行安装动作前应预配置《~/.bashrc》环境变量声明。

可通过创建“命令脚本+systemd 服务”的方式实现 Synopsys License 认证服务的开机自启动运行, 为此须留意解除因 AlmaLinux 默认启用 SELinux 导致的文件操作权限受限问题。

演示了各项 Synopsys 软件的启动运行方式, 确认 License 授权与软件运行环境依赖均

已完成部署，其中为 VCS 的启动验证制作了极简测试用例《vcs_smoke》。

最后利用 AI 生成了十款 Synopsys 软件的自动化部署 Bash (Shell) 脚本，并在 CentOS 7.9 中进行了实测确认。该脚本可适用于 AlmaLinux、CentOS 7 和 Ubuntu 三类 Linux 系统，支持十款 Synopsys 软件的批量安装配置与卸载。

须特别提出的是，Synopsys 软件编译尤其在跑大型 ASIC 设计时非常吃电脑内存，有可能需要数百 GB 内存，应综合考虑自身应用情况予以足额配备。

环境搭好，接下来就是一个个工具一步步流程，好好学习了。

千里之行，始于足下。老刘的炼芯之路（ASIC 全栈开发），才刚刚开始~

【萧伯纳语录】

世上有所成就的人，都是主动寻找自己想要的环境；如果找不到，就自己创造出来。

The people who get on in this world are the people who get up and look for the circumstances they want, and, if they can't find them, make them.

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